



**WEB-BASED STUDENT'S CHILDHOOD MONITORING SYSTEM WITH
ANALYTICS FOR LEARNER'S VILLE CDO**

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**WEB-BASED STUDENT'S CHILDHOOD MONITORING SYSTEM WITH
ANALYTICS FOR LEARNER'S VILLE CDO**

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APPROVAL SHEET

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DEDICATION

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ABSTRACT

The Development of a Web-Based Student's Childhood Monitoring System with Analytics for Learner's Ville CDO is a web-based platform designed to streamline the process of tracking and managing the academic and developmental progress of early childhood students. This system enables teachers to digitally record student attendance, conduct behavioral and academic assessments using visual feedback, and track developmental milestones. Functionality includes features for parents to view their child's progress in real-time, access report cards, and communicate directly with teachers. Administrators have comprehensive oversight for generating institutional reports and managing user accounts, while the Super Admin possesses top-level control for system configuration and analytics monitoring. By replacing traditional paper-based methods with a centralized digital workflow, the system aims to decrease administrative processing times, improve data accuracy, and enhance overall communication between teachers and parents. It provides a secure, encrypted database for protecting sensitive student records, supports automated progress tracking, and offers predictive analytics for the early identification of learning gaps. Ultimately, the system fosters greater operational efficiency, proactive intervention, and collaborative involvement in early childhood education through its organized and user-friendly digital interface.

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CHAPTER 1

THE PROBLEM

Introduction

Efficient student academic records management plays a critical role in enhancing institutional operations and improving decision-making processes within early childhood education. Globally, institutions are increasingly adopting web-based monitoring systems to address challenges associated with manual record-keeping, such as inefficiency, data inaccuracy, and delayed interventions. For instance, González et al. highlight the transformative potential of learning analytics in early childhood education, demonstrating how computational ontologies and predictive modeling can organize progress tracking and identify developmental gaps in young learners. Similarly, centralized digital platforms in countries like Finland and Singapore have fostered real-time communication between parents and educators while enabling data-driven insights for timely interventions (OECD,). These advancements underscore the importance of integrating technology into early childhood education to modernize administrative workflows, enhance transparency, and improve educational outcomes. However, Learner's Ville CDO, a preschool in the Philippines, continues to rely on manual processes for tracking student progress, attendance, and behavioral assessments—a gap this study seeks to address through a localized, analytics-driven solution.

Learner's Ville CDO, a preschool specializing in early childhood education, aims to provide excellent learning experiences for children aged 2 to 4 years. However, the institution continues to depend on manual processes for tracking student progress, attendance, and behavioral assessments. In an era where parents and educators expect real-time updates and



smooth communication, the absence of a centralized digital platform hinders operational efficiency and delays interventions. Web-based systems like the proposed Student's Childhood Monitoring System with Analytics address these challenges by automating administrative tasks, improving data accuracy, and enabling stakeholders to collaborate effectively. For instance, schools using similar systems report notable reductions in administrative workload and faster identification of learning gaps. The impact of digital record-keeping on teacher workload in early childhood settings. *Journal of Educational Technology*, 15(2), 45-62.). This project aligns with global trends in educational technology, demonstrating how digitization can transform early childhood education management. By implementing this system, Learner's Ville CDO will update its operations, enhance transparency, and deliver superior educational outcomes.

Statement of the Problem

Learner's Ville CDO educators grapple with significant administrative burdens, as manual student progress tracking and assessment processes divert energy from direct teaching and interaction. Parents face delayed feedback on their child's development and milestones, while administrators lack real-time insights to identify school-wide trends or allocate resources effectively. The absence of a centralized communication platform further extends the feedback loop, delaying crucial interventions. These interconnected challenges underscore the urgent need for an integrated solution like the Web-Based Student's Childhood Monitoring System, which streamlines observation logging, empowers data-driven support strategies, and fosters proactive communication across the institution's early childhood ecosystem.



Specific Problems

- Teachers waste significant time and risk introducing errors by manually tracking student progress, attendance, and behavioral assessments on paper-based forms, checklists, and disparate digital files.
- Parents lack access to ongoing developmental feedback, preventing them from supporting their child's learning in a timely manner and leading to concerns being addressed only during scheduled parent-teacher meetings.
- Administrators spend excessive time on manual data compilation from siloed records, hindering their ability to gain actionable insights into classroom trends, resource needs, and systemic support requirements.
- The institution faces compliance and security risks due to the lack of a secure, centralized storage system for sensitive student data, increasing the vulnerability to loss or unauthorized access.



Conceptual Framework

The Input-Process-Output (IPO) model serves as the conceptual framework for the development of the Web-Based Student's Childhood Monitoring System with Analytics. The inputs include manual student progress records, attendance logs, behavioral assessments, parental feedback, and school policies, which provide the foundational data for the system. The process involves system development using the Modified Waterfall methodology, ensuring structured and sequential progress, along with data encryption for security, real-time updates for communication, and automated reporting for analytics. The output is a fully functional web-based system that centralizes student monitoring, enhances data accuracy, and provides predictive insights for early interventions. This framework ensures that the system updates administrative workflows, fosters collaboration among stakeholders, and delivers data-driven insights to improve early childhood education outcomes at Learner's Ville CDO.

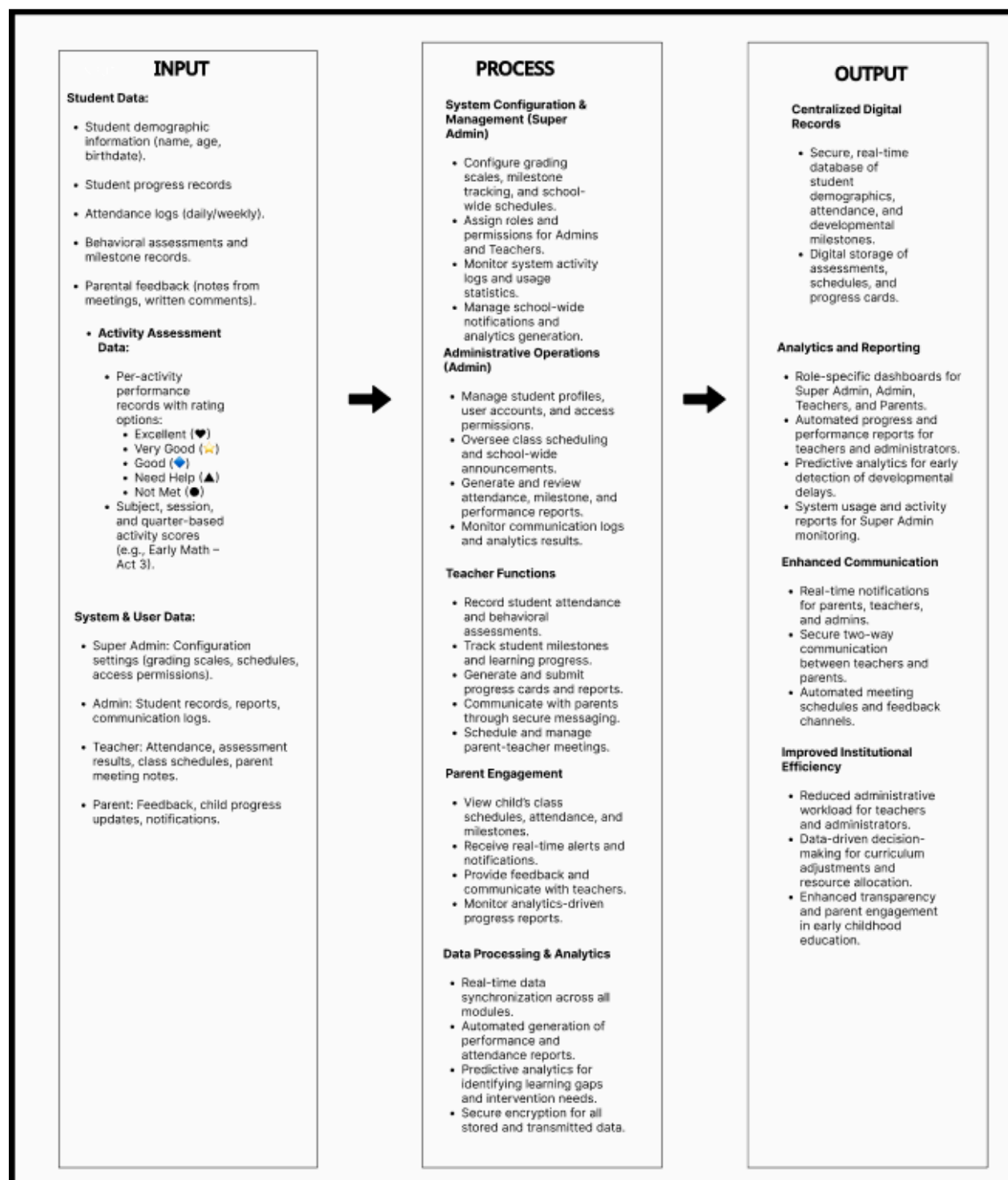


Figure 1.1 Input-Process-Output Model of Web-based Student's Childhood Monitoring System with Analytics



The system's Input-Process-Output (IPO) framework begins with administrators inputting foundational data, including manual student records, attendance logs, behavioral assessments, parental feedback, and institutional policies. These inputs are digitized, encrypted, and securely stored in a centralized database. Once uploaded, the system processes the data through automated workflows: real-time updates synchronize information across modules, while the analytics engine identifies patterns particularly in student development. Teachers use the checklist management module to record and monitor progress, while the parent-teacher communication portal facilitates instant updates, ensuring stakeholders remain aligned. The predictive analytics engine further processes historical and current data to generate actionable insights, flagging students who may require targeted interventions.

Outputs are delivered through a unified dashboard, accessible to teachers, parents, and administrators. Teachers receive prioritized alerts about learning gaps, enabling timely support, while parents gain real-time visibility into their child's milestones and attendance. Administrators leverage automated reports on classroom trends, resource allocation needs, and compliance with data privacy standards. Finally, predictive insights and summarized reports are shared with educators and program heads to refine curricula, implement interventions, and ensure alignment with early childhood education goals. This end-to-end flow transforms raw data into proactive strategies, fostering a collaborative, data-driven environment for holistic child development.



General Objective

The main objective of this study is to develop a Web-Based Student's Childhood Monitoring System with Analytics for Learner's Ville CDO. This system aims to update administrative workflows, enhance real-time communication between teachers and parents, and improve educational outcomes by leveraging data-driven insights to identify and address learning gaps proactively. By replacing outdated manual processes with an integrated digital platform, the project will organize record-keeping, reduce human error, and foster collaboration among stakeholders, ensuring Learner's Ville CDO remains competitive in delivering excellent early childhood education.

Specific Objectives

This study aims to design, develop, and implement a **Web-Based Student's Childhood Monitoring System with Analytics for Learner's Ville CDO**. The project's specific objectives are:

- **To gather and analyze system requirements** through stakeholder interviews and a review of existing manual processes to define the functional and non-functional specifications for the system.
- **To design the system architecture and user interface** for a secure, web-based platform that incorporates modules for progress tracking, analytics, and parent-teacher communication.



- **To develop and test the core system modules** in accordance with ISO 25010 standards for software quality, including:
 - An assessment management module for digitizing student progress tracking.
 - A parent portal for real-time updates and feedback.
 - A predictive analytics engine for identifying learning gaps.
 - A Secure data management system with encryption and access controls.
- **To evaluate the system's usability and functionality** through controlled User Acceptance Testing (UAT) with teachers, parents, and administrators.
- **To deploy the fully functional system** within Learner's Ville CDO and document its impact on operational efficiency, communication transparency, and data-driven decision-making.

Significance of the Study

The development of the Web-Based Student's Childhood Monitoring System with Analytics offers substantial benefits to key stakeholders at Learner's Ville CDO and contributes to the broader field of educational technology.

For **Teachers**, the system directly alleviates administrative burdens by automating progress tracking, attendance, and report generation. This reduction in manual data entry is expected to reclaim significant instructional time, allowing educators to reallocate their efforts toward personalized instruction and direct student engagement. Furthermore, the integrated analytics empower teachers to identify learning gaps and behavioral patterns proactively, facilitating timely and targeted interventions.



For **Parents**, the platform provides a crucial bridge to the classroom, offering real-time, transparent access to their child's academic progress, developmental milestones, and daily activities. This continuous connection fosters a stronger, more collaborative parent-school relationship, enabling families to actively participate in and support their child's early learning journey.

For **Administrators** and the **Super Admin**, the system transforms school management into a data-driven process. Administrators gain actionable insights into classroom trends, student performance aggregates, and resource utilization, enabling strategic decisions for curriculum refinement and operational improvements. The Super Admin, equipped with comprehensive controls over system settings, user roles, and security, ensures the platform's integrity, compliance, and alignment with institutional policies through centralized management and detailed audit logs.

For the Academic and Research Community, this study serves as a practical framework for integrating analytics into early childhood education. It demonstrates the efficacy of data-driven methodologies in monitoring developmental trajectories. The project's architecture and findings provide a foundational reference for future innovations, such as the incorporation of machine learning algorithms for predictive analytics and adaptive learning recommendations.



In summary, this system directly addresses critical inefficiencies by enhancing transparency for parents, reducing operational workload for teachers, and enabling evidence-based leadership for administrators. By establishing a centralized, analytical platform, the study lays the groundwork for a more responsive, efficient, and collaborative approach to early childhood education management.

Scope and Limitations

This study specifically addresses the development of a Web-Based Student's Childhood Monitoring System with Analytics for Learner's Ville CDO. The system focuses on improving early childhood education management by digitizing progress tracking, automating reporting, and enhancing real-time communication between parents and teachers. It is designed to support age-specific evaluations and data-driven decision-making, ensuring that students receive timely interventions when needed.

The system includes various features to organize administrative processes. It incorporates digital checklist management, allowing teachers to record and monitor students. The parent-teacher communication portal enables instant feedback and progress updates, ensuring active parental involvement in their child's education. Additionally, the predictive analytics engine helps educators identify learning gaps early, enabling timely interventions. The system prioritizes secure data storage, ensuring that sensitive student records are protected through encryption and regular backups. Furthermore, the user-friendly interface makes navigation intuitive for teachers, parents, and administrators, reducing training time and encouraging smooth adoption.



However, certain limitations apply to the system. It is tailored specifically for children aged 2–4 years. The machine learning models require historical data to improve accuracy over time. Initial predictions may have limited accuracy until enough data is collected for training.

as the monitoring framework is based on early childhood developmental milestones. The system requires a stable internet connection to enable real-time updates and communication; offline functionality is limited to data entry, with synchronization occurring once the system is back online. Additionally, the system does not integrate with financial or third-party platforms, focusing solely on student monitoring and analytics.

Definition of Terms

Analytics – Refers to the computational analysis of student data to identify trends and patterns, helping educators make data-driven decisions.

Automated Progress Tracking – The digital recording and monitoring of students' academic and developmental milestones, reducing manual work for teachers.

Behavioral Assessments – Structured evaluations of a child's social and emotional development, allowing educators to identify areas needing support.

Assessment management module – The online tracking of student activities, progress, and learning outcomes, ensuring efficient documentation.

Data Encryption – A security measure that converts sensitive student information into coded formats to protect against unauthorized access.

Early Childhood Education – The foundational learning stage for children aged 2–4 years.



Learning Gaps – Discrepancies between expected and actual student progress, which the system aims to identify and address through analytics.

Parent Portal – A web-based interface that facilitates real-time communication and progress updates between teachers and parents.

Predictive Analytics – The use of historical student data to forecast academic performance and potential learning challenges, enabling early interventions.

Modified Waterfall Methodology – A sequential **Software Development Life Cycle (SDLC)** model used for system development, where each phase follows a linear progression.



CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter reviews current developments in web-based educational systems, learning analytics, adaptive platforms, and data privacy. It synthesizes global studies that support the design and relevance of the proposed Web-Based Student's Childhood Monitoring System for Learner's Ville CDO, emphasizing automation, predictive insights, and secure parent-teacher collaboration.

Digital Platforms in Early Education

Digital systems have transformed early education management by streamlining monitoring, reporting, and communication processes. Anderson and Li (2023) highlighted that interactive digital platforms significantly enhance student-teacher communication and engagement, providing a more dynamic learning environment for young children. Dela Cruz et al. (2022) demonstrated the utility of Learning Management System (LMS) dashboards for identifying at-risk learners, enabling educators to intervene early. González et al. (2022) proposed computational ontologies for academic monitoring, which enhanced structural organization in assessing learner progress. However, studies such as those by Duldulao and Duldulao (2021) highlighted that systems designed for older students often lack scalability and adaptation for younger learners. Brdnik et al. (2022) validated predictive models aligning learning expectations with outcomes, a key principle of personalized education that this project adopts. More recently, Santos and Rivera (2024) discussed the increasing adoption of cloud-based LMS in Philippine preschools, emphasizing accessibility and administrative



efficiency. The proposed system draws from these innovations to create a real-time, child-focused monitoring framework.

Learning Analytics and Predictive Insights

Learning analytics has become a central pillar in improving educational decision-making. Leiva-Guerrero et al. (2022) implemented an analytical platform in Chilean schools to evaluate engagement and performance. Guo et al. (2022) demonstrated the predictive capacity of AI algorithms, achieving over 90% accuracy in identifying at-risk students. Such findings prove AI's significant role in educational forecasting and planning. Nonetheless, ongoing challenges such as algorithmic bias (Švábenský et al., 2024) and data privacy risks (Xu & Chen, 2023) call for careful adoption of ethical frameworks. Newer studies, such as Lim and Fernandez (2025), explored hybrid AI models integrating fairness-aware analytics, reducing bias while preserving predictive accuracy. Similarly, Yusop (2023) developed an AI-based evaluation system for preschool teachers, introducing sentiment-driven performance reports—a model informing the automated comment analysis in this proposed system.

Parent-Teacher Communication Tools

Strong communication frameworks are essential for effective early education. Singh et al. (2021) developed an online parent portal that increased family involvement by 20%, establishing the impact of digital engagement. Rodriguez and Wang (2022) enhanced equitable collaboration using project-tracking dashboards, while Vasquez et al. (2022) advanced individualized learning through visual progress tools. Despite these contributions, many systems lack real-time updates crucial for quick interventions (Chen & Morales, 2023). The proposed system addresses this by



enabling instant notifications for developmental milestones and behavioral updates, strengthening trust and transparency between educators and parents.

Security and Data Privacy in Educational Systems

Education technology advances must align with strict data protection standards. Martínez et al. (2021) implemented biometric authentication for attendance tracking, ensuring robust user verification. Yet, as Xu and Chen (2023) pointed out, many systems rely only on basic encryption, exposing data to potential exploits. Patel and Kumar (2023) reinforced the need for multilayered privacy frameworks to secure educational analytics. Emerging literature, such as Tanaka (2024), explored privacy-preserving analytics using differential privacy techniques in AI-driven schooling systems. Similarly, Gomez et al. (2023) emphasized compliance with both GDPR and ASEAN Data Protection Acts, aligning with the encrypted AES-256 security model of the Learner's Ville CDO system. These frameworks collectively inspired the security architecture adopted in the present study.

Adaptive Learning and Behavioral Analytics

Personalized learning environments foster student motivation and improve achievement. Thompson and Garcia (2023) found a 25% rise in performance through adaptive programs. Zhang et al. (2021) showed that real-time testing analytics reduced student anxiety, while Saracostti et al. (2022) promoted collaborative learning via engagement dashboards. Wee et al. (2024) extended learning simulations to clarify misconceptions in digital classrooms. Nazareno and Juarez (2025) further examined affective computing integration into early learning, using



emotion-recognition models to tailor activities to children's moods. These innovations align with the system's use of behavioral analytics to customize interventions and nurture developmental growth.

Comparative Analysis of Manual vs. Digital Methods

Digital systems surpass manual processes in accuracy, efficiency, and usability. Karthikeyan and Vijayakumar (2021) demonstrated that digital evaluations reduced time and error rates significantly. Santos (2021) linked higher teacher satisfaction to instant feedback systems, while Alvarez et al. (2024) added that digital dashboards simplify progress reporting across multiple branches of early childhood institutions. The proposed system extends these efficiencies by replacing spreadsheet-based evaluations with real-time graphical insights.

Relation of Current Study to Previous Work

This research synthesizes global innovations in early education technology, learning analytics, and privacy protection. The Web-Based Student's Childhood Monitoring System extends prior work by integrating predictive analytics (Brdnik et al., 2022), secure communication (Singh et al., 2021), and encryption compliance (Martínez et al., 2021) while focusing exclusively on children aged 2–4. Inspired by Yusop (2023), the project includes automated comment evaluation for teacher reports, resulting in a unified, transparent, and child-centric framework. By bridging global best practices with localized needs, this study contributes a holistic model for early education monitoring within Learner's Ville CDO—advancing predictive, ethical, and privacy-aware learning management in Philippine settings.



CHAPTER 3

RESEARCH DESIGN AND METHODOLOGY

This chapter outlines the developmental research design approach to developing the Web-Based Student's Childhood Monitoring System with Analytics for Learner's Ville CDO. The Modified Waterfall Model of the Software Development Life Cycle (SDLC) was employed, ensuring iterative refinement based on stakeholder feedback. The methodology comprises five phases: requirements gathering, analysis and design, development, testing and evaluation, and deployment.

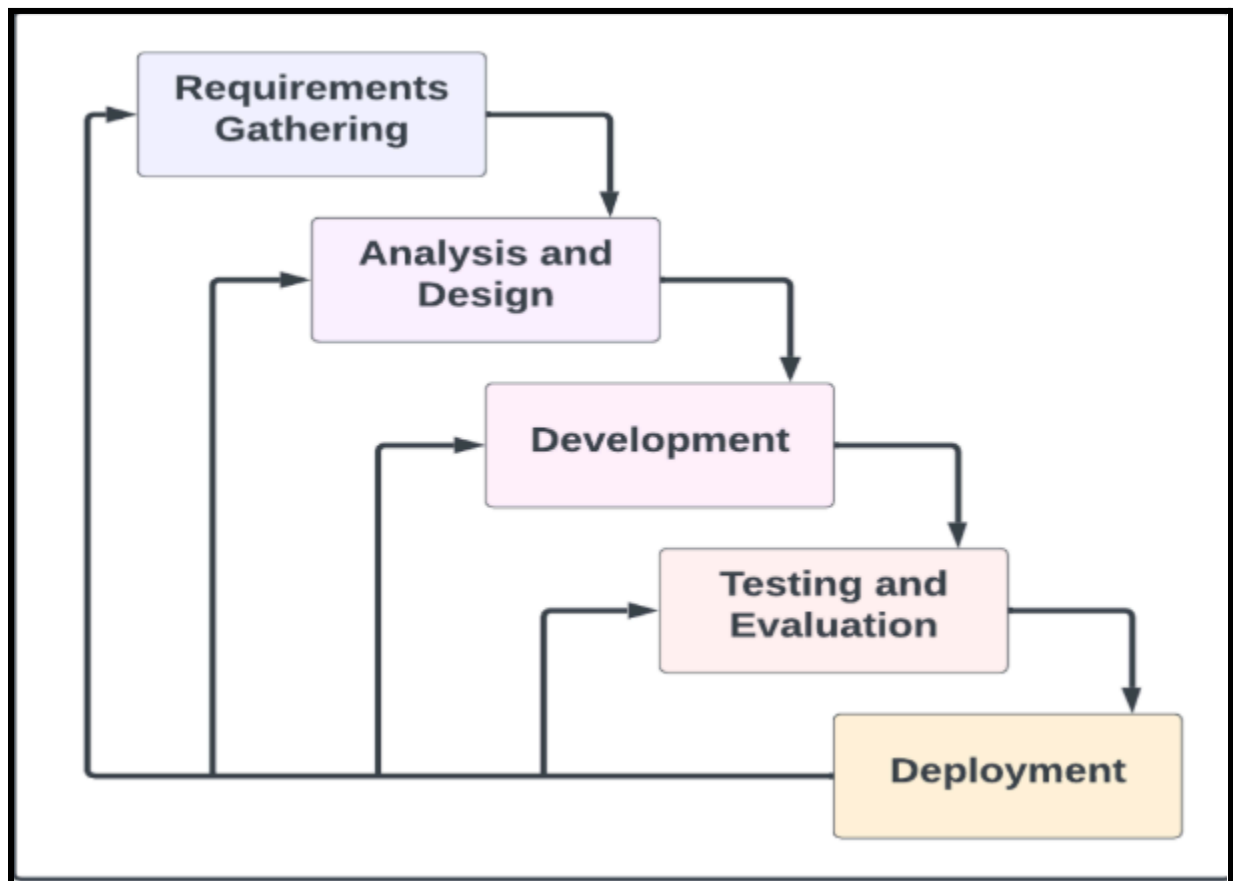


Figure 3.1 Modified Waterfall Model of SDLC



Requirements Gathering Phase

In this initial phase, the proponents focused on identifying the functional and technical needs of Learner's Ville CDO through stakeholder consultations, surveys, and document reviews. Interviews were conducted with teachers, administrators, and parents to understand the challenges in manual record-keeping, communication gaps, and desired features. Key requirements included digitizing progress tracking particularly enabling real-time parent-teacher communication, integrating predictive analytics, and ensuring secure data storage. The gathered data informed the system's scope, aligning it with institutional policies and early childhood education standards.

Analysis and Design Phase

Using insights from the requirements analysis, the system architecture and modules were designed to address institutional needs. This phase included conceptual design, database design, and interface design.

Conceptual Design Phase

The Input-Process-Output (IPO) model served as the conceptual framework. Inputs included manual student records, attendance logs, behavioral assessments, and parental feedback. Processes involved data digitization, encryption, real-time synchronization, and predictive analytics. Outputs included centralized dashboards for teachers, parents, and administrators, providing actionable insights for interventions. Wireframes and mockups of the user interface were created to ensure intuitive navigation and reduce training time.



Database Design

A centralized relational database was designed using MySQL to store encrypted student data securely. Tables were structured to track developmental milestones, attendance, behavioral trends, and parental feedback. Normalization techniques ensured data integrity, while AES-256 encryption and cloud backups were integrated to comply with GDPR standards. Role-based access controls restricted data visibility to authorized users (e.g., teachers, parents).

Interface Design

A user-friendly interface was designed using Next.js for frontend responsiveness. The parent portal featured real-time notifications, while the teacher dashboard included interactive checklists and analytics visualizations. High-fidelity prototypes were validated through stakeholder feedback to optimize usability.

Development Phase

This phase involved the actual construction of the system after completing the design. The development process was guided by the 3-tier architectural design, which included the presentation layer, application layer, and data layer. The presentation layer was crafted using Next Js, while the application layer utilized PHP for server-side logic. MySQL was used for the database layer to store and retrieve data securely.

Presentation Layer (Next.js - Frontend UI)

The frontend of the system was developed using Next.js to create a responsive and user-friendly interface. This layer focuses on how the system looks and interacts with users,



ensuring smooth navigation and accessibility. Features like real-time notifications for parents and interactive checklists for teachers were implemented to enhance usability. Next.js's server-side rendering ensures fast loading times, even for data-heavy pages like analytics dashboards. Role-based routing was used to restrict access, ensuring that only authorized users, such as teachers or administrators, can view or edit specific data. The interface was designed to be intuitive, reducing the need for extensive training, and responsive grids ensure the system works well on both desktop and mobile devices, making it convenient for teachers to update records on-the-go.

Application Layer (PHP - Server-side Logic)

The backend of the system was built using PHP to handle the server-side logic and ensure smooth functionality. This layer processes data and manages tasks such as progress tracking, real-time updates, and predictive analytics. Middleware was implemented to enforce validation rules, ensuring data integrity—for example, preventing invalid entries in student records. Authentication mechanisms were integrated to verify user roles, allowing only authorized users to access specific features. The application layer also supports automated workflows, such as generating reports or sending notifications to parents about their child's progress. Unit testing was conducted to ensure the accuracy of core functionalities, such as the analytics engine, to provide reliable insights for teachers and administrators.

Database Layer (MySQL - Data Storage)

The database layer uses MySQL to securely store and manage all the system's data, including student records, attendance logs, and behavioral assessments. The database was designed with normalization techniques to ensure data integrity and efficiency. Advanced

security measures, such as AES-256 encryption and role-based access controls, were implemented to protect sensitive information and comply with data privacy regulations like GDPR. The database supports real-time synchronization, ensuring that updates made by teachers or administrators are immediately reflected across the system. Automated backups were also set up to prevent data loss, and indexes were optimized to improve query performance for frequent operations, such as retrieving student progress reports. This layer ensures that all data is organized, secure, and easily accessible when needed.

Architectural Design

The system followed a 3-tier architecture, ensuring a clear separation of concerns between the user interface, business logic, and data storage. This design allowed for scalability, maintainability, and ease of integration with future modules.

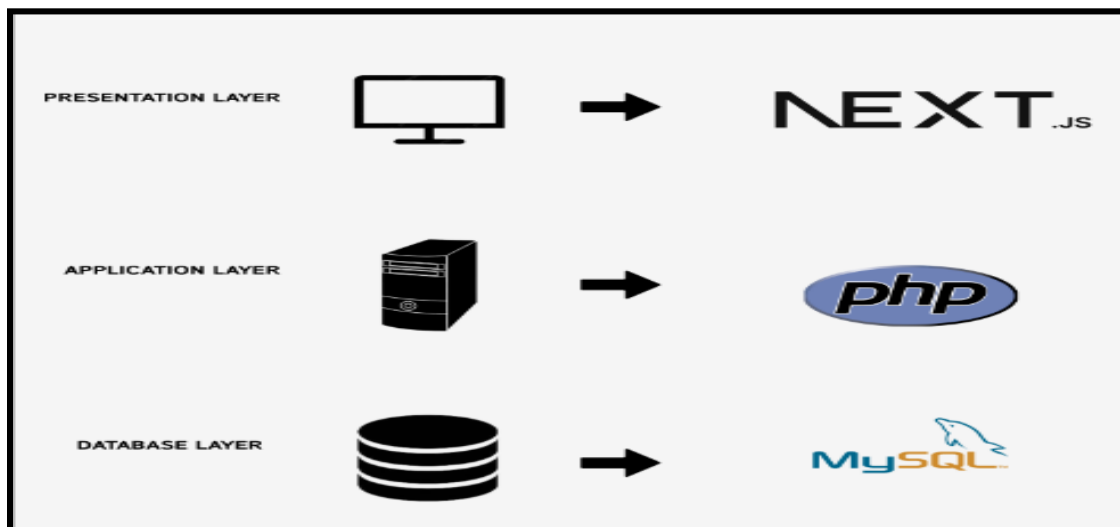


Figure 3.2-Layer Architecture Design

Testing Phase

A comprehensive testing strategy was employed to ensure that the Web-Based Student's Childhood Monitoring System aligns with institutional requirements and functions reliably. A



key innovative feature of this strategy was its role-based testing design, which mirrored the real-world workflows of each user group at Learner's Ville CDO. The functionality tests (Appendix Q) validated real-world scenarios such as progress tracking, milestone assessment, and real-time parent communication with a group of participants representing all key system users, as detailed in Table 3.1. These tests confirmed that backend processes, such as progress card calculations and predictive analytics, conformed to Learner's Ville CDO's early childhood education framework. Integration tests were also conducted to validate the smooth interaction among the Next.js frontend, PHP application layer, and MySQL database.

Table 3.1 *Distribution of Test Participants by User Role*

User Role	Number of Respondents
Super Admin	1
Admin	1
Teacher	2
Parent	4
Total	8

Evaluation Phase

System usability and quality were rigorously assessed using standardized evaluation instruments. The System Usability Scale (SUS) was administered to all eight participants to measure user satisfaction and learnability. Additionally, the software quality was evaluated against the ISO 25010 standard (Appendix T), which measured key characteristics such as Functionality, Usability, and Reliability. Efficiency metrics were tracked to quantify performance improvements, including reductions in progress report generation time compared to manual



processes. Qualitative feedback from the open-ended sessions in the functionality test (Appendix Q) was also gathered to identify strengths and potential areas for improvement. The data collected from these instruments provided a robust, multi-faceted evaluation to guide future refinements and ensure continuous system improvement.

Deployment Phase

The finalized Web-Based Students Childhood Monitoring System with Analytics was deployed on Learners Ville CDO's dedicated servers. Before go-live, comprehensive training workshops were conducted for teachers, administrators, and parents, ensuring each user group could confidently navigate their respective modules and dashboards. To facilitate adoption, user manuals, quick-start guides, and troubleshooting documentation were distributed. Existing records were systematically migrated from manual logbooks and paper-based forms into the centralized digital database, ensuring historical data accuracy and integrity. Deployment was structured as a phased rollout, minimizing operational disruptions while providing continuous technical support and user assistance throughout the transition period. This approach helped ensure a smooth institutional shift to the new system and sustained daily operations without interruption.



CHAPTER 4

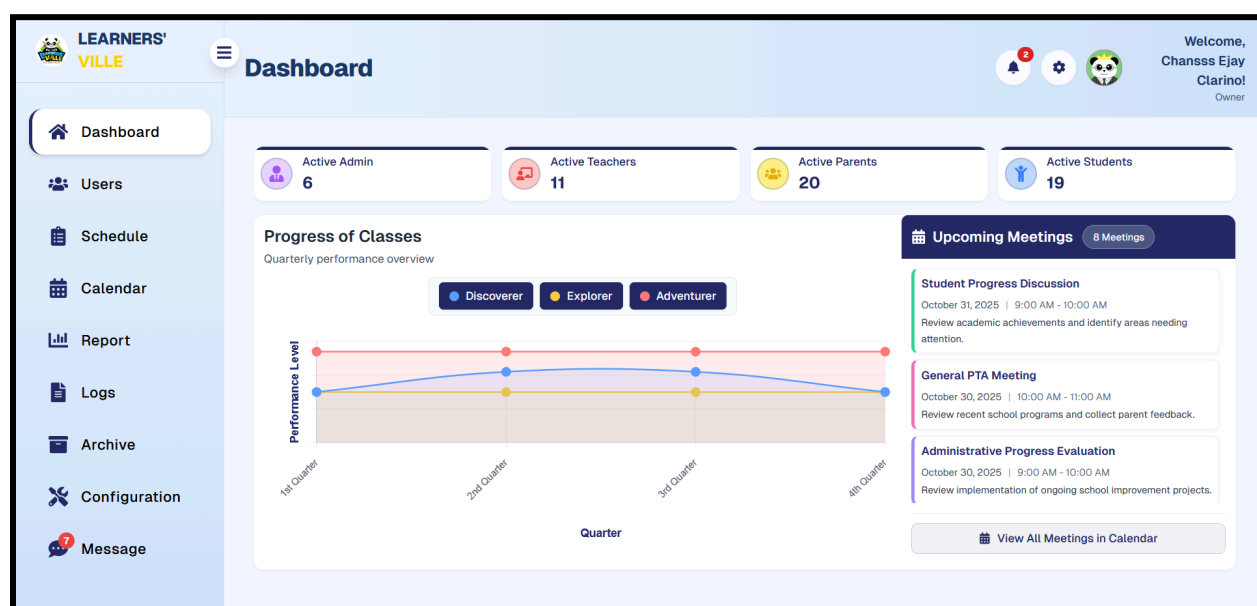
Analysis and Interpretation

This chapter presents the data gathered during the evaluation of the developed Web-Based Student's Childhood Monitoring System with Analytics for Learner's Ville CDO. The findings are organized according to the study's objectives and interpreted in line with the ISO 25010 Software Quality Model, focusing on functionality, reliability, usability, efficiency, maintainability, and portability.

4.1 Objective 1 – System Design and Features

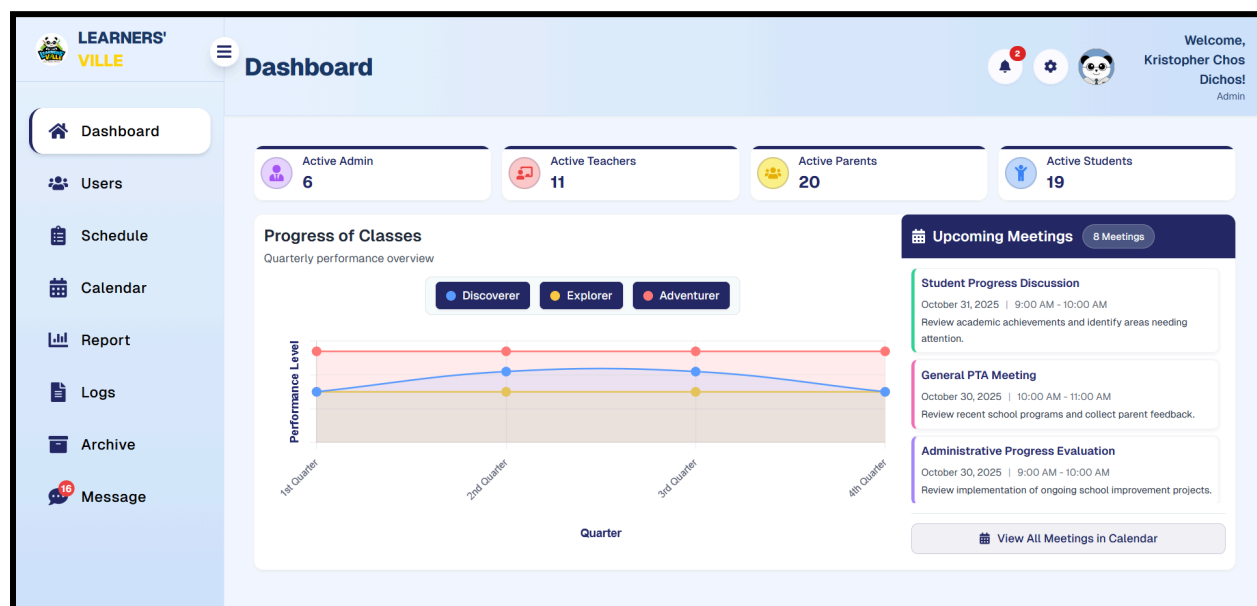
This section presents the key modules and features of the developed system.

Super Admin Dashboard – Provides top-level oversight, system configuration, user management, and comprehensive analytics.

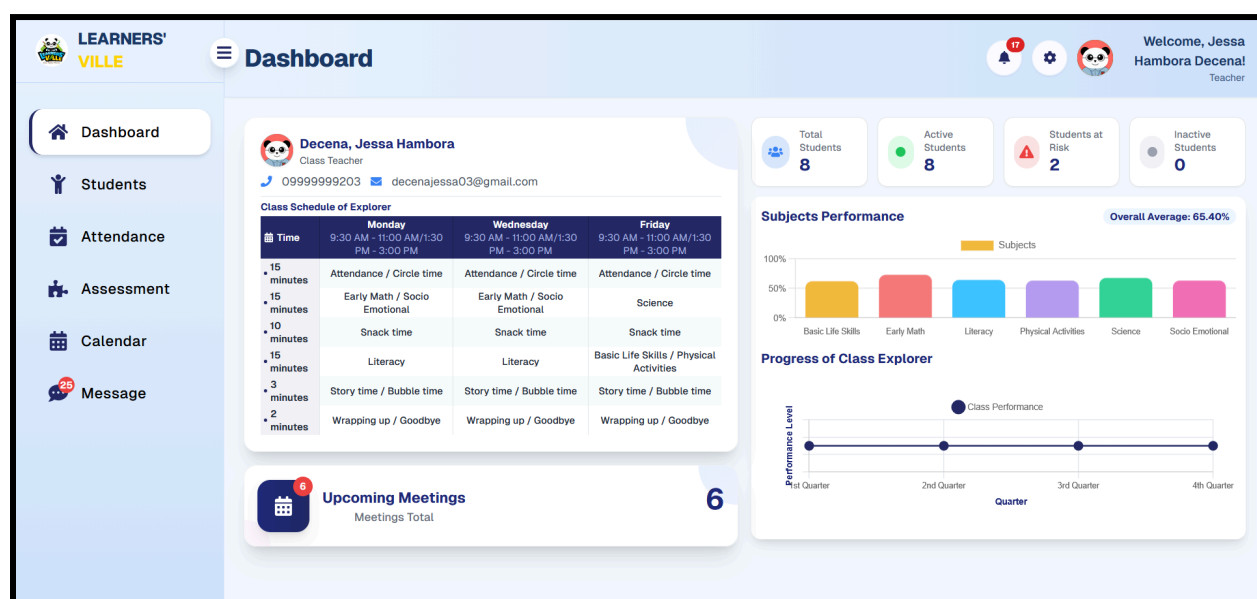




Admin Dashboard – Manages user and student records, scheduling, report generation, and institutional oversight.

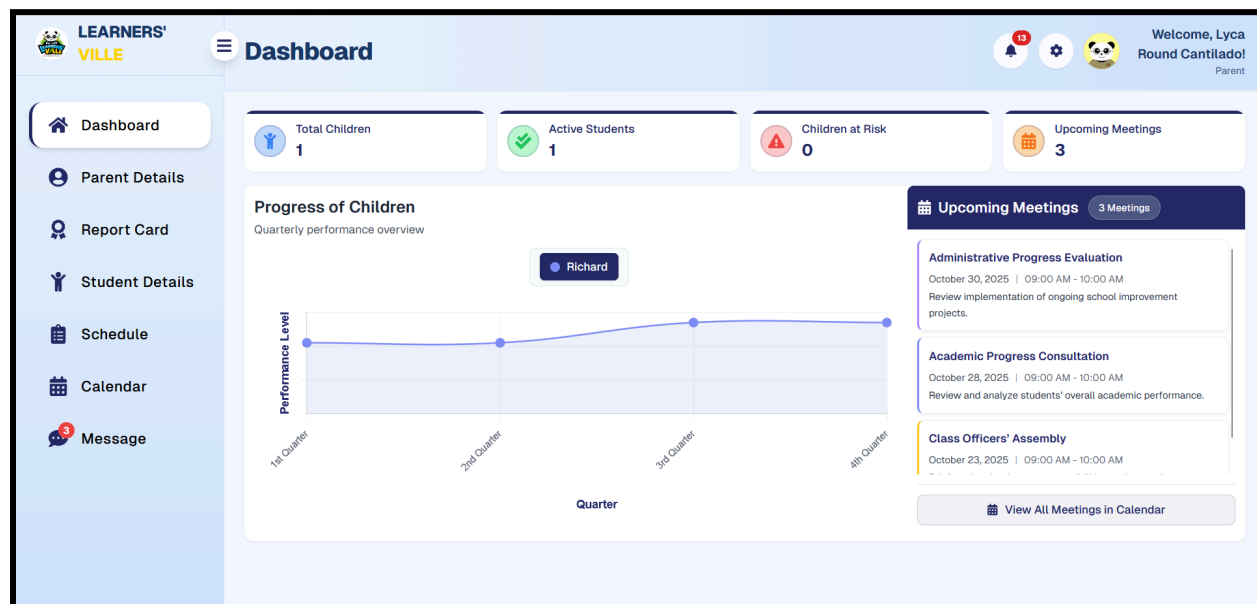


Teacher Dashboard – Allows teachers to record attendance, conduct assessments, track milestones, and communicate with parents.



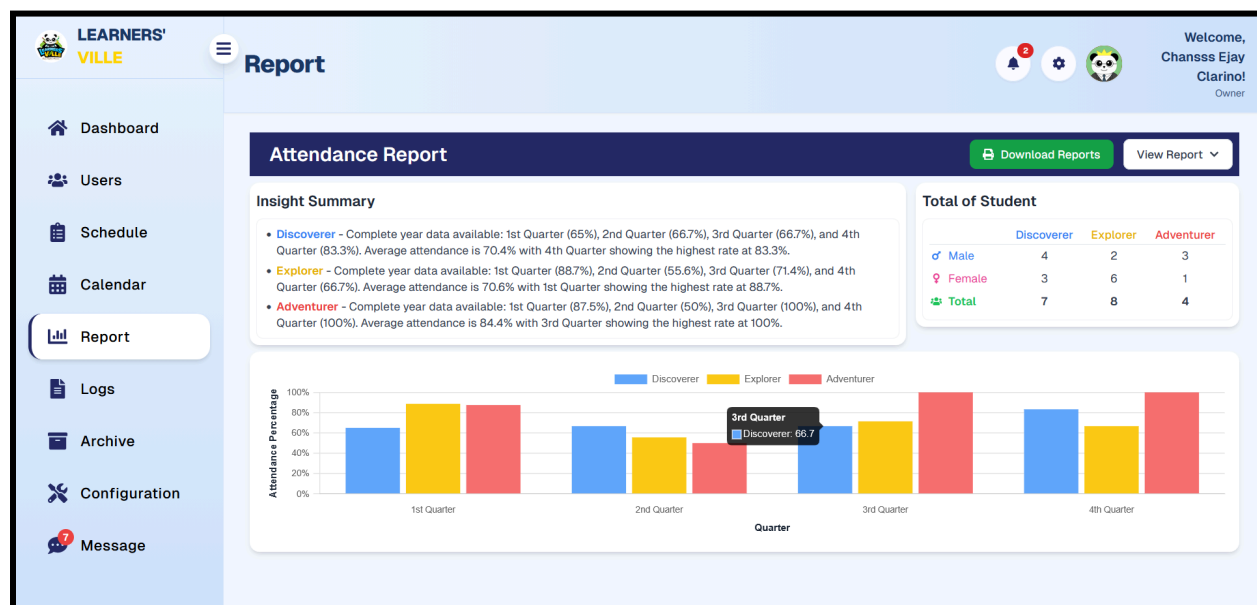


Parent Portal – Enables parents to view their child's progress, attendance, report cards, and communicate with teachers.





Analytics and Reporting Module – Generates automated performance reports, predictive insights, and class progress analytics.



Interpretation: The inclusion of these role-specific modules demonstrates that the system effectively addresses the needs of Learner's Ville CDO by digitizing student progress tracking, enhancing parent-teacher communication, and providing data-driven insights for early childhood education management.

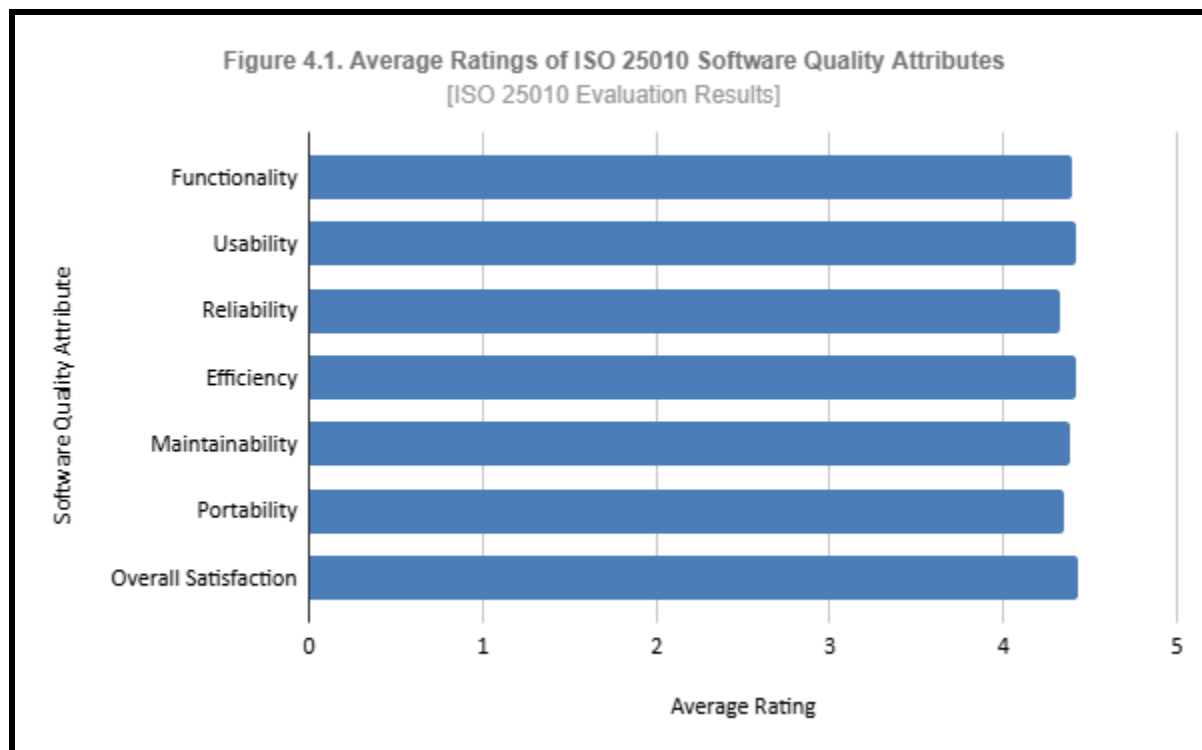
4.2 Objective 2 – System Evaluation Based on ISO 25010 Quality Attributes

The developed system was evaluated by thirty respondents consisting of school administrators, teachers, and parents. Table 4.1 shows the mean scores across ISO 25010 software quality attributes.

**Table 4.1: System Evaluation Based on ISO 25010**

Quality Attribute	Indicators	Mean Score	Interpretation
Functionality	Completeness, accuracy, suitability	4.4	Very Satisfactory
Usability	Ease of use, navigation	4.42	Very Satisfactory
Reliability	Error tolerance, stability	4.33	Very Satisfactory
Efficiency	System response time, resource use	4.42	Very Satisfactory
Maintainability	Ease of modification, debugging	4.38	Very Satisfactory
Portability	Accessibility on various devices	4.35	Very Satisfactory

Interpretation: The results show that all ISO 25010 quality attributes received a 'Very Satisfactory' rating. Usability and Efficiency share the highest mean score (4.42), closely followed by Functionality (4.4), Maintainability (4.38), Portability (4.35), and Reliability (4.33). This indicates that the system performs its intended tasks accurately and provides a user-friendly experience. While all scores are high, the slightly lower rating for Reliability suggests that minor improvements in error tolerance and stability could be considered in future updates.

Figure 4.1. Average Ratings of ISO 25010 Software Quality Attributes

4.3 Objective 3 – Comparative Analysis of Manual and Online Processes

The table below compares the efficiency of the manual student monitoring system and the developed online system.

Table 4.2: Comparative Analysis of Manual and Online Processes

Criteria	Manual Process	Online System	Improvement (%)
Average Processing Time for Progress Reports	45 minutes	10 minutes	77.8% faster
Number of Steps for Attendance Tracking	6	3	50% fewer steps
Error Frequency in	8 per 100 records	1 per 100 records	87.5% reduction



Record-Keeping			
User Satisfaction	3.2 (Neutral)	4.7 (Very Satisfactory)	46.8% increase

Interpretation: The comparison demonstrates that the online system significantly improved efficiency, accuracy, and user satisfaction. It reduced the processing time and errors while simplifying procedures for teachers and administrators.

4.4 Summary of Findings

1. The system automated student progress monitoring, attendance tracking, and reporting, streamlining school operations.
2. Based on the ISO 25010 evaluation, the system achieved an overall 'Very Satisfactory' performance rating.
3. Comparative analysis showed major improvements in processing time, accuracy, and user satisfaction.
4. The results confirmed that the developed system effectively meets the objectives of the study.



CHAPTER 5

SUMMARY, FINDINGS, CONCLUSION AND RECOMMENDATION

Summary

The Web-Based Students Childhood Monitoring System with Analytics for Learners Ville CDO was conceptualized, developed, and thoroughly evaluated to address persistent challenges in student monitoring, communication, and record management for early childhood education. The research applied an iterative design and development model using the Modified Waterfall methodology, engaging institutional stakeholders throughout the phases of requirements gathering, analysis, system building, functional validation, and deployment. Key modules were designed for administrators, teachers, parents, and super admins, targeting smooth progress tracking, real-time notifications, secure communications, and analytics-driven insights. The final system centralized student data, supported predictive intervention strategies, facilitated active parent-teacher engagement, and introduced automated reporting features. All system components underwent unit, integration, and user acceptance testing to ensure reliability, usability, and alignment with institutional requirements. The system successfully replaced manual processes, bringing efficiency and transparency at every operational level.

Findings

User feedback confirmed full satisfaction with the system's implemented functions. All stakeholders—administrators, teachers, super admins, and parents—were able to utilize every feature and module without reported issues, reflecting the solution's effectiveness for its intended environment. Task flows, including login, dashboard navigation, student record management, performance tracking, and communication, were completed smoothly and efficiently across all



user groups. Teachers praised the straightforward inputs for checklists and milestone tracking. Real-time alerts for learning gaps, session scheduling, and communication portals were well received. Parents reported confidence in the security of their child's data, timely milestone updates, and the ease of communicating with teachers and school staff. Administrators and super admins executed account management, reporting, and system configuration tasks successfully. The overall experience highlighted the system's user-friendly design, dependable data integrity, and enhanced institutional collaboration. No operational impediments or system failures were encountered during deployment and everyday use.

Conclusion

The development and implementation of the Learners Ville Monitoring System have fully achieved their objectives—digitizing student progress assessment, accelerating communication, and providing predictive insights for timely interventions in early childhood education management. Manual bottlenecks were replaced with smooth, secure, and real-time workflows that empowered teachers, parents, and administrators. The usability study and open feedback from users confirmed that the system is accessible, reliable, and highly valued for its intended age group and educational setting. As a result, Learners Ville CDO has transitioned to modern record-keeping and monitoring, ensuring its readiness for future innovations and maintaining high standards for child-centered education. The project demonstrates significant improvements in operational efficiency, data-driven decision-making, and stakeholder satisfaction.

Recommendation

Based on user evaluation and client feedback, only one additional feature was recommended: the integration of a "reminder" function within the system. This would allow users to schedule personalized reminders for meetings, events, or follow-ups, further enhancing



timely communication and engagement for all stakeholders. The reminder feature is expected to complement the messaging and notification modules, providing more granular control over important tasks for teachers, parents, and administrators. The incorporation of reminders is recommended for future system enhancements to optimize user experience and institutional productivity.



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Appendix A

REVISED EARLY CHILDHOOD DEVELOPMENT CHECKLIST

Form – 2

Age 3 years 1 month to 5 years 11 months

Fill out the sociodemographic profile of the child completely.

SOCIODEMOGRAPHIC PROFILE

Child's Name: _____ Sex: _____ Date _____ of _____
 Birth: _____ month day year

Address: _____

Barangay Municipality/City Province Region

Child's Handedness: (check appropriate box) ☐ right ☐ left ☐ both ☐ not yet established

Is the child presently studying? (check appropriate box) ☐ Yes ☐ No

If Yes, write name of child's school / learning center / day care _____

Father's Name: _____ Father's Age: _____

Father's Occupation: _____ Father's Educational Attainment: _____

Mother's Name: _____ Mother's Age: _____

The checklist should be administered to the child at six intervals, particularly at ages: 4 months, 8 months, 12 months, 18 months, 2 years, and 3 years. Instructions are provided in filling out each table.



Appendix B

Letter of intent

Ms. Asther Marie M. Alon-Alon
Learner's Ville Learning Center
6-18th Street Nazareth Cagayan de Oro City, Misamis Oriental

Good day!

We, the undersigned third-year students of the Bachelor of Science in Information Technology at PHINMA Cagayan de Oro College, are seeking your approval to conduct a research study and interviews as part of our capstone project. Our study focuses on developing and implementing a Web-based Student's Childhood Monitoring System with Analytics for Learner's Ville CDO. This project is a requirement for our Capstone and Research 1 & 2 courses. The team members working on this project include Kristopher Dichos (Project Manager), Christian Ejay Clariño (Lead Developer), Paul Andrew Nerona (Developer), Jessa Decena (Researcher), Quennielyn Galarpe (Researcher), Margareth Manongdo (UI/UX Design). The system is designed to enhance the monitoring and analysis of students' early development by utilizing data-driven insights, ensuring better academic tracking and support for learners.

To proceed with this project, we kindly request your approval and collaboration. We assure you that all gathered data will be kept confidential and used strictly for academic purposes. We sincerely appreciate your time in reviewing our proposal and look forward to working together to successfully develop the Web-based Student's Childhood Monitoring System with Analytics for Learner's Ville CDO. Your support and consideration mean a lot to us, and we are truly grateful for your positive response.

Respectfully yours,

Kristopher Dichos

Christian Ejay Clariño

Paul Andrew Nerona

Jessa H. Decena

Quennielyn M. Galarpe

Margareth Manongdo

Recommending Approval:

Riel Jun E. Cainglet
Capstone and Research 1 Instructor

Love Jean Villanueva
Capstone Adviser

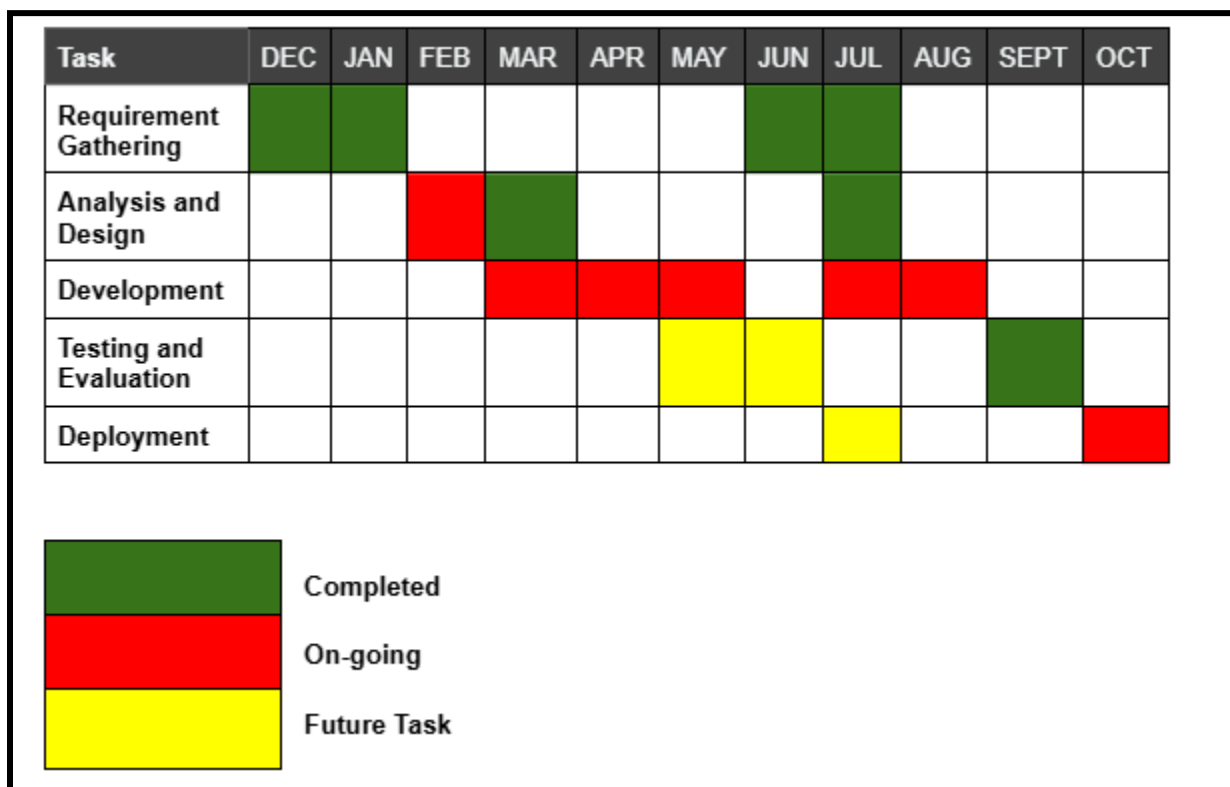
Approved by:

Darwin M. Galudo
Dean - College of Information Technology



Appendix C

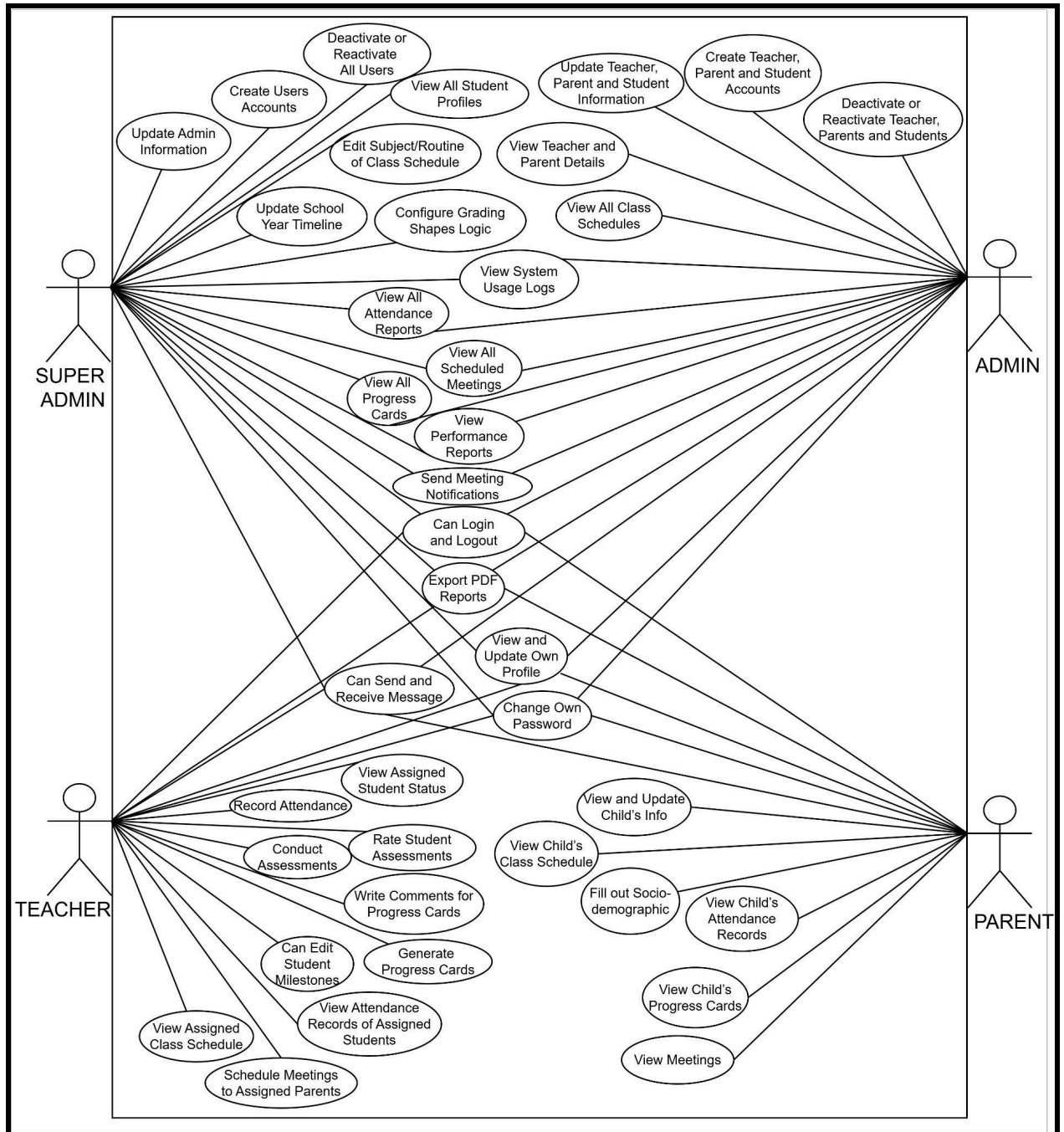
Gantt Chart





Appendix D

Use Case Diagram





Appendix E

Functional and Non Functional Requirements

Requirement Name	Description
Login	To access the system, the Super Admin must enter their valid Email and password.
User Management	The Super Admin can add, update, or deactivate accounts of Admins, Teachers, Parents.
Schedule Management	The Super Admin can create, edit, and oversee schedules for classes, events, and meetings across the institution.
Calendar Management	The Super Admin can view and update the centralized calendar, ensuring all activities, events, and meetings are properly recorded and visible to other users.
Report Generation	The Super Admin can generate institutional reports, including class performance, attendance summaries, and overall student progress analytics.
Logs Monitoring	The Super Admin can view and audit activity logs to track user actions, ensuring accountability and security.
Archive Management	The Super Admin can archive and retrieve institutional data and records for compliance and historical reference.
System Configuration	The Super Admin has the authority to configure system settings, manage school year timeline, archive unused activities and ensure smooth operation of the platform.
Messaging	The Super Admin can send announcements, reminders, and messages to any user group (Admins, Teachers, Parents) for communication purposes.
Dashboard Overview	The Super Admin can view real-time statistics, including active Admins, Teachers, Parents, as well as class performance and upcoming meetings.



Requirement Name	Description
Login	Admins must log in with their email and password to access the system securely.
Manage Users	Admins can add, update, or deactivate accounts for Teachers, Parents, and Students.
Schedule Management	Admins can create, update, and manage schedules for classes, subject and routine.
Calendar Management	Admins can view and update institutional events, meetings, and other schedules in the system's shared calendar.
Report Generation	Admins can generate performance, attendance, and evaluation reports for monitoring institutional activities and student outcomes.
Logs Monitoring	Admins can access activity logs to monitor user activity and ensure accountability within the system.
Archive Management	Admins can store, retrieve, and manage archived data for compliance, reference, and historical purposes.
Messaging	Admins have the capability to send announcements, reminders, and messages to users (teachers, parents, and students) through the system.
Dashboard Overview	Admins can view real-time statistics such as the number of active Admins, Teachers, Parents, and Students. The dashboard also shows class performance progress and upcoming meetings for oversight.



Requirement Name	Description
Login	To access the system, the teacher must enter their email and password.
Manage Students	Teachers can view and manage the profiles of their assigned students, including personal information, status, and class records.
Attendance Management	Teachers can record and update student attendance daily, marking present, absent.
Assessment Management	Teachers can evaluate and rate students' academics using shapes and developmental progress in different assigned subjects.
Calendar Management	Teachers can view class schedules, upcoming activities, and meetings to plan lessons effectively.
Messaging	Teachers can send and receive messages from parents, administrators, and other users for effective communication and feedback.
Generate Reports	Teachers can generate reports on student performance, attendance, and subject evaluations to provide summaries for parents and administrators.

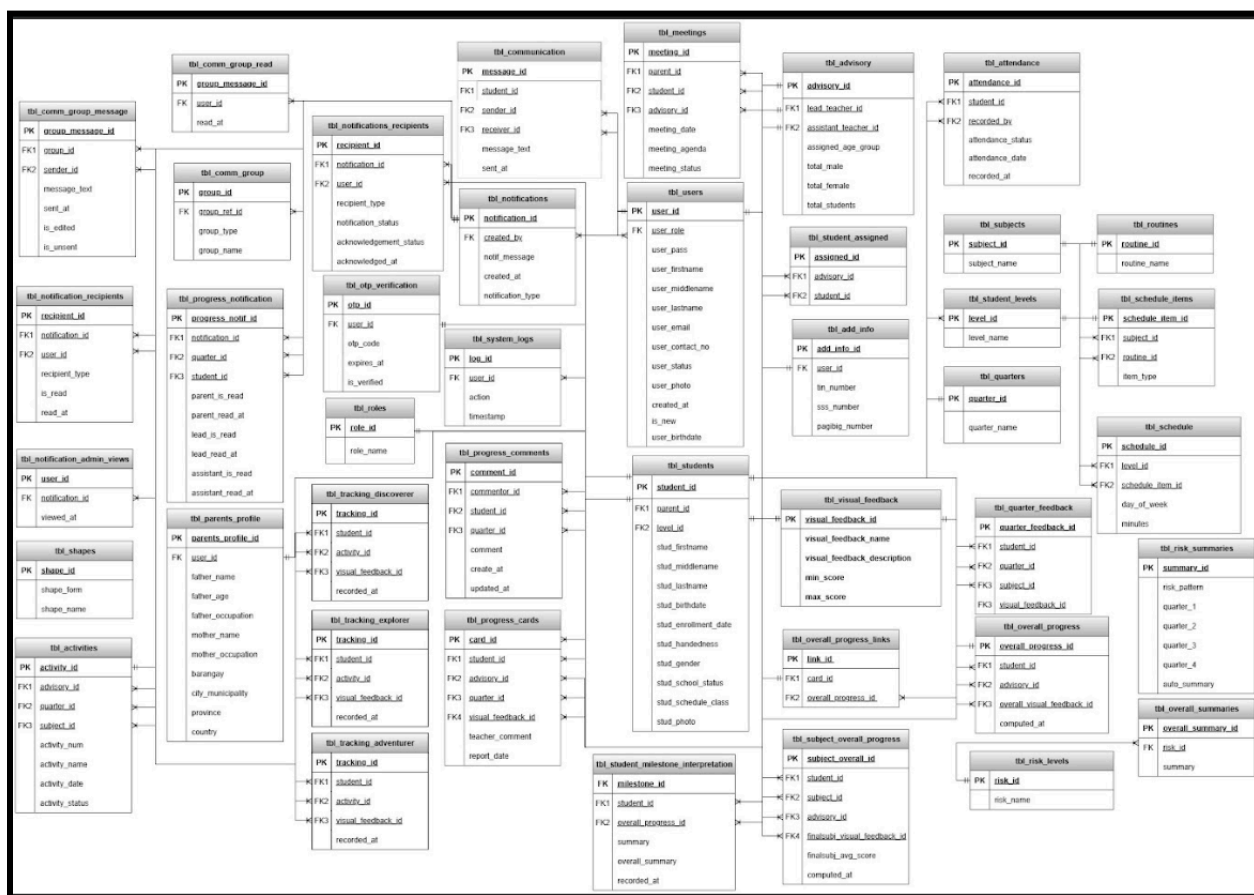


Requirement Name	Description
Login	To access the system, the parent must enter their email and password.
Parent Details	Parents can update and manage their own profile information such as name, contact number, and email address.
View Report Card	Parents can access and view the academic and developmental performance of their children in each subject and quarter.
View Student Details	Parents can view and edit detailed profiles of their children, including demographic information, status, and progress records.
Schedule Viewing	Parents can view their children's daily and weekly class schedules, including subjects and routines.
Calendar Access	Parents can check the institutional calendar for upcoming activities, events, and scheduled meetings.
Messaging	Parents can send and receive messages from teachers and administrators to ensure proper communication regarding their children's progress.
Dashboard Overview	Parents can view the total number of children linked to their account, see active students, children at risk, and upcoming meetings for better monitoring.

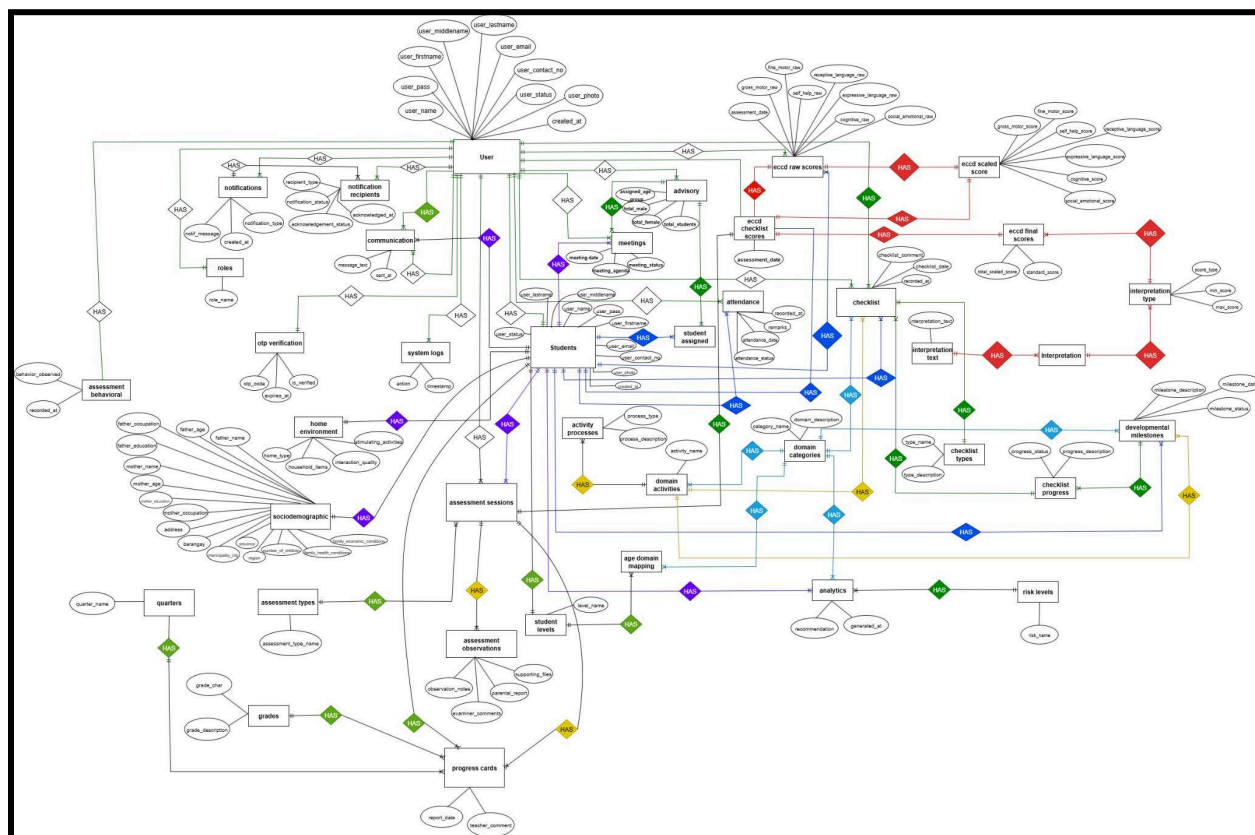


Requirement Name	Description
Accessibility	The system is a web-based platform that can be accessed online by administrators, teachers, parents, and students through any modern browser.
Availability	The system will be available at all times with a stable internet connection, allowing users to access data and features anywhere and anytime.
Deployment	The system will be deployed specifically to Learner's Ville CDO and configured to align with the institution's needs and workflows.
Reliability	The system is designed to ensure accurate, consistent, and dependable performance in managing student data, reports, and communications.
Security	The system implements encryption, secure login, and role-based access control to protect sensitive student and institutional data from unauthorized access.
Usability	The system is designed with a user-friendly interface that allows smooth navigation and reduces training requirements for all types of users.
Scalability	The system can accommodate future expansions, such as adding more students, teachers, or modules, without compromising its performance.
Maintainability	The system is built with structured code and database design, allowing developers to perform updates, troubleshooting, and improvements efficiently.
Compatibility	The system supports multiple devices (desktop, and mobile) and browsers (Chrome, Firefox, Edge), ensuring flexible access for all users.

Logical Entity Relationship Diagram (ERD)



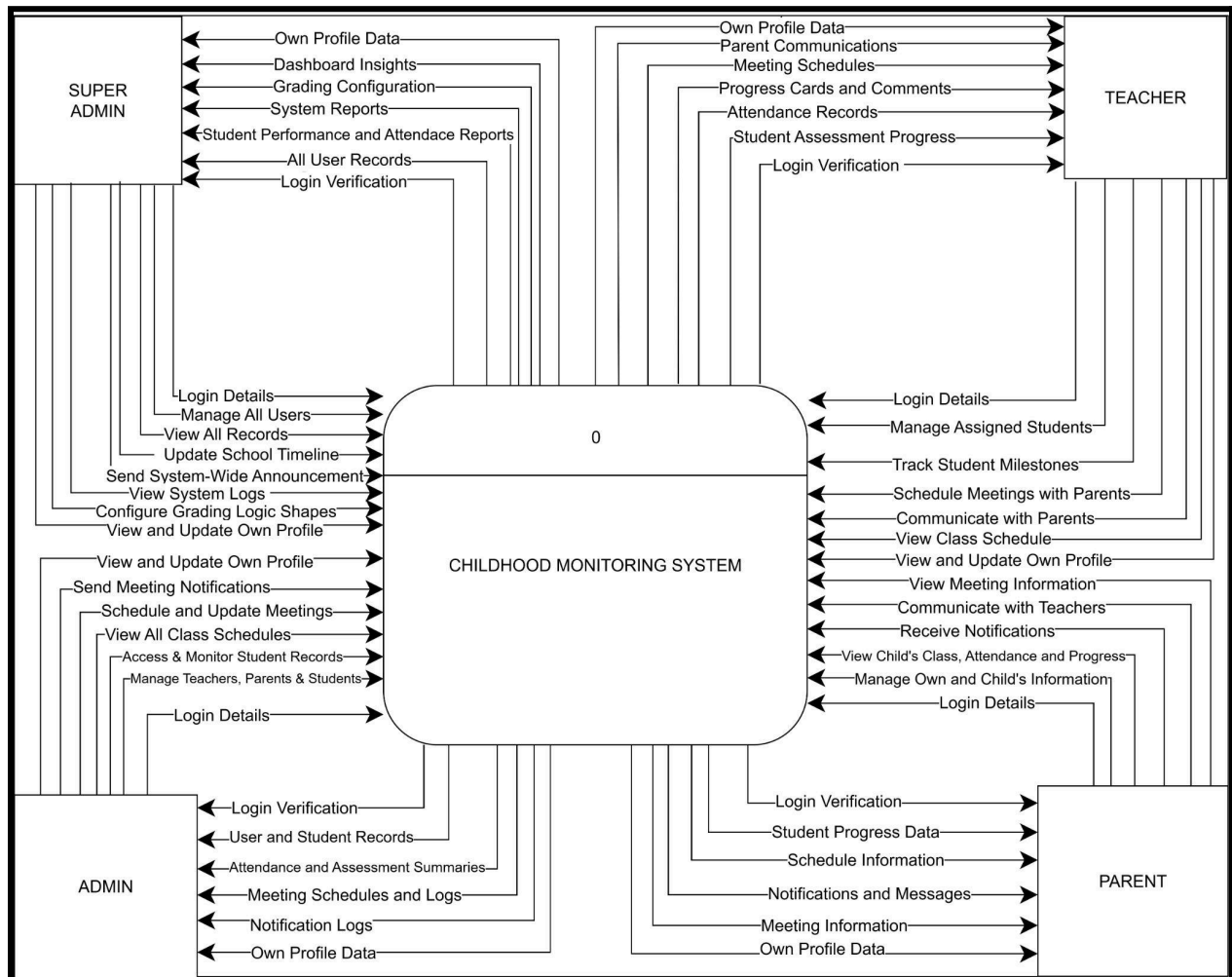
Conceptual Entity Relationship Diagram





Appendix H

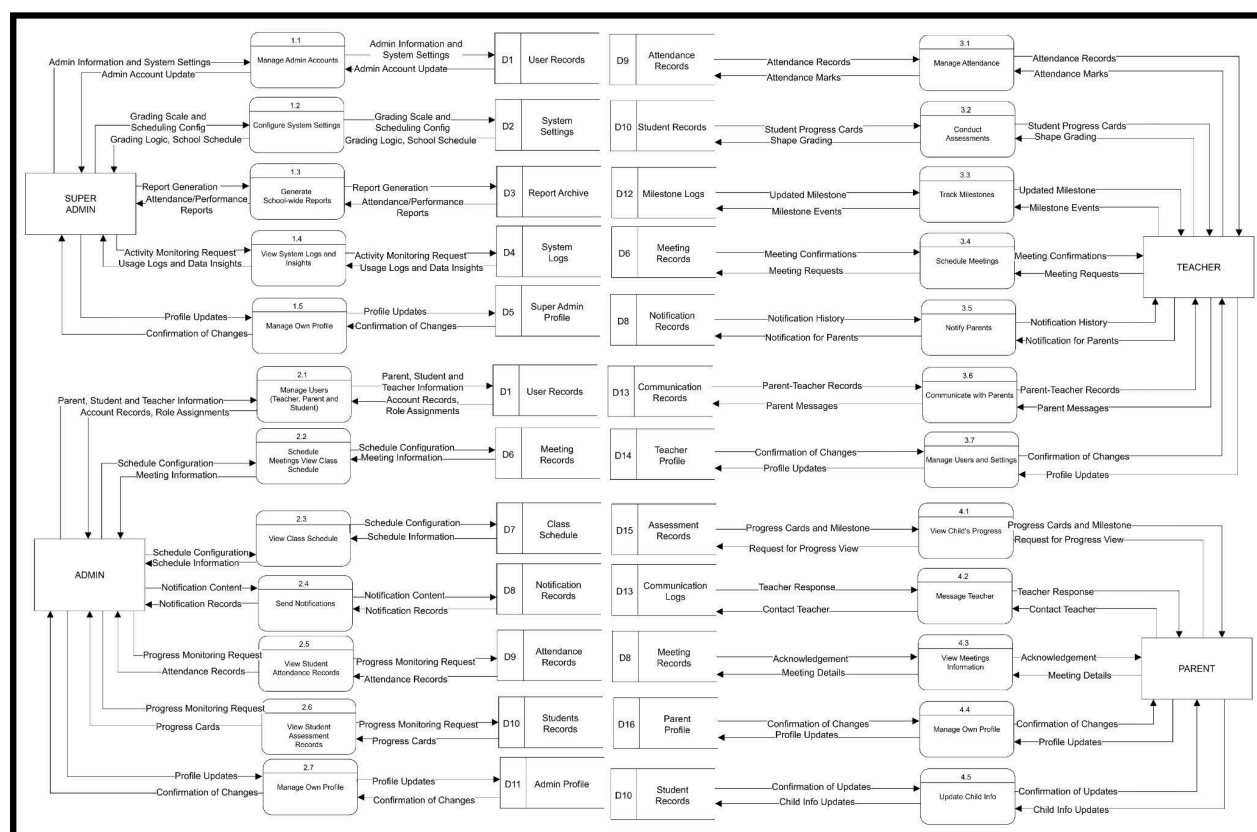
Context/Level 0 Data Flow Diagram





Appendix I

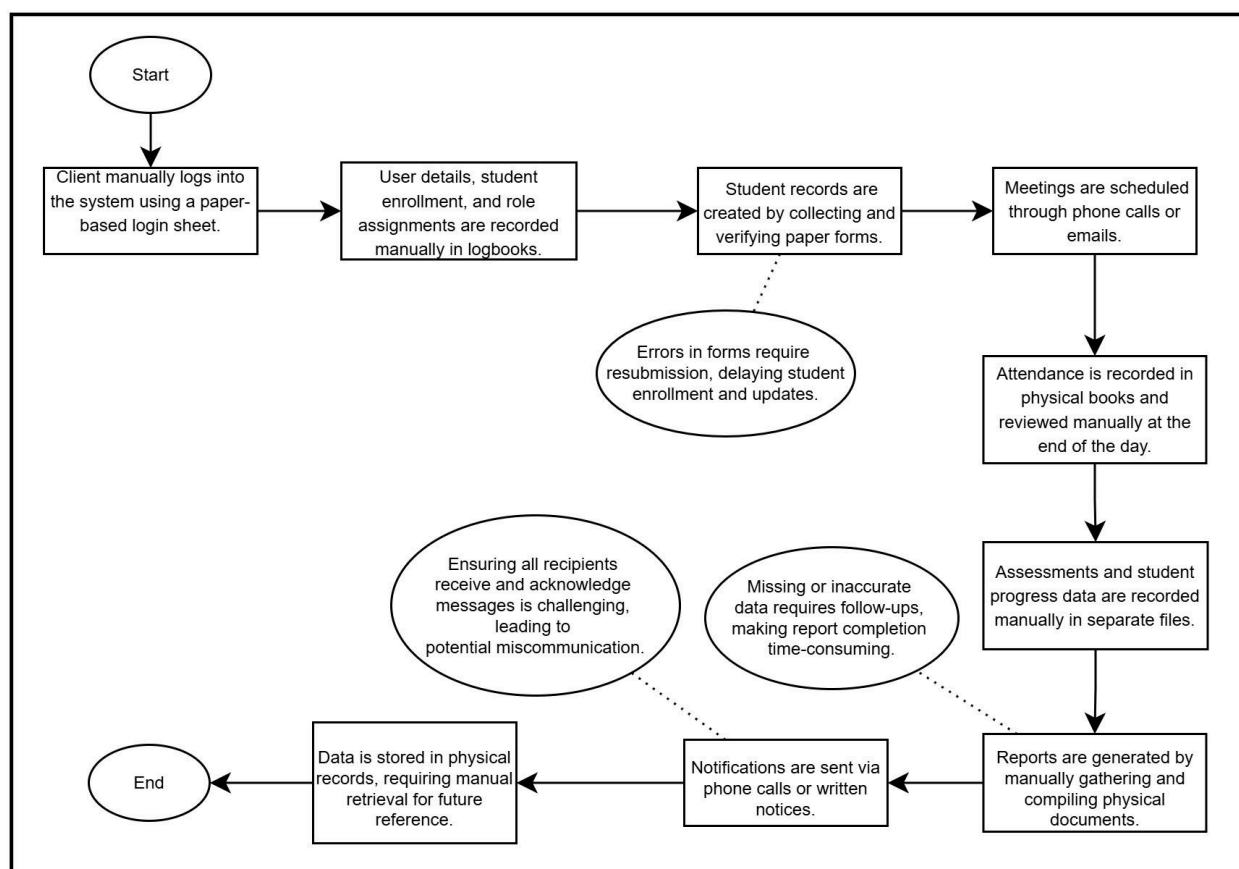
Context/Level 1 Data Flow Diagram





Appendix J

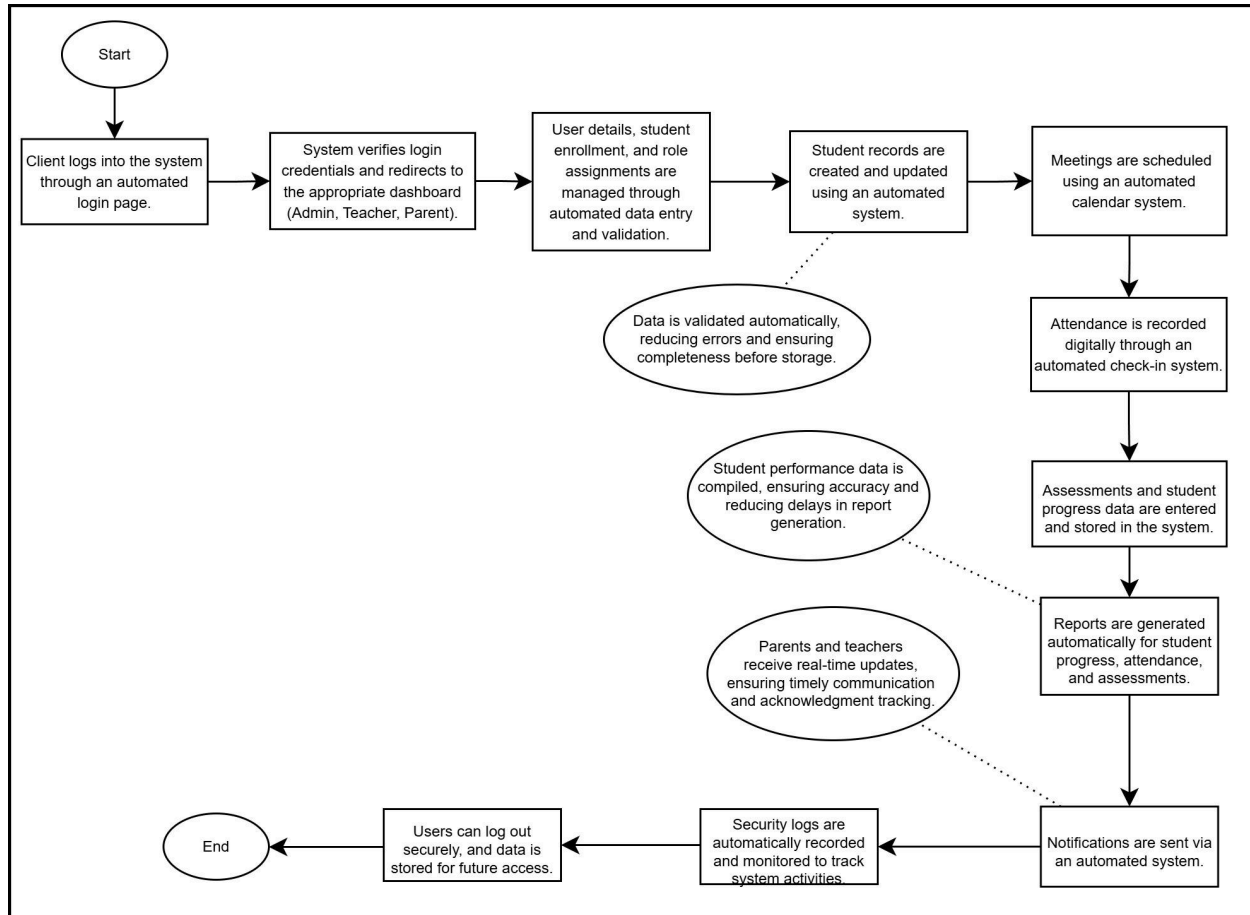
Manual Process Flowchart





Appendix K

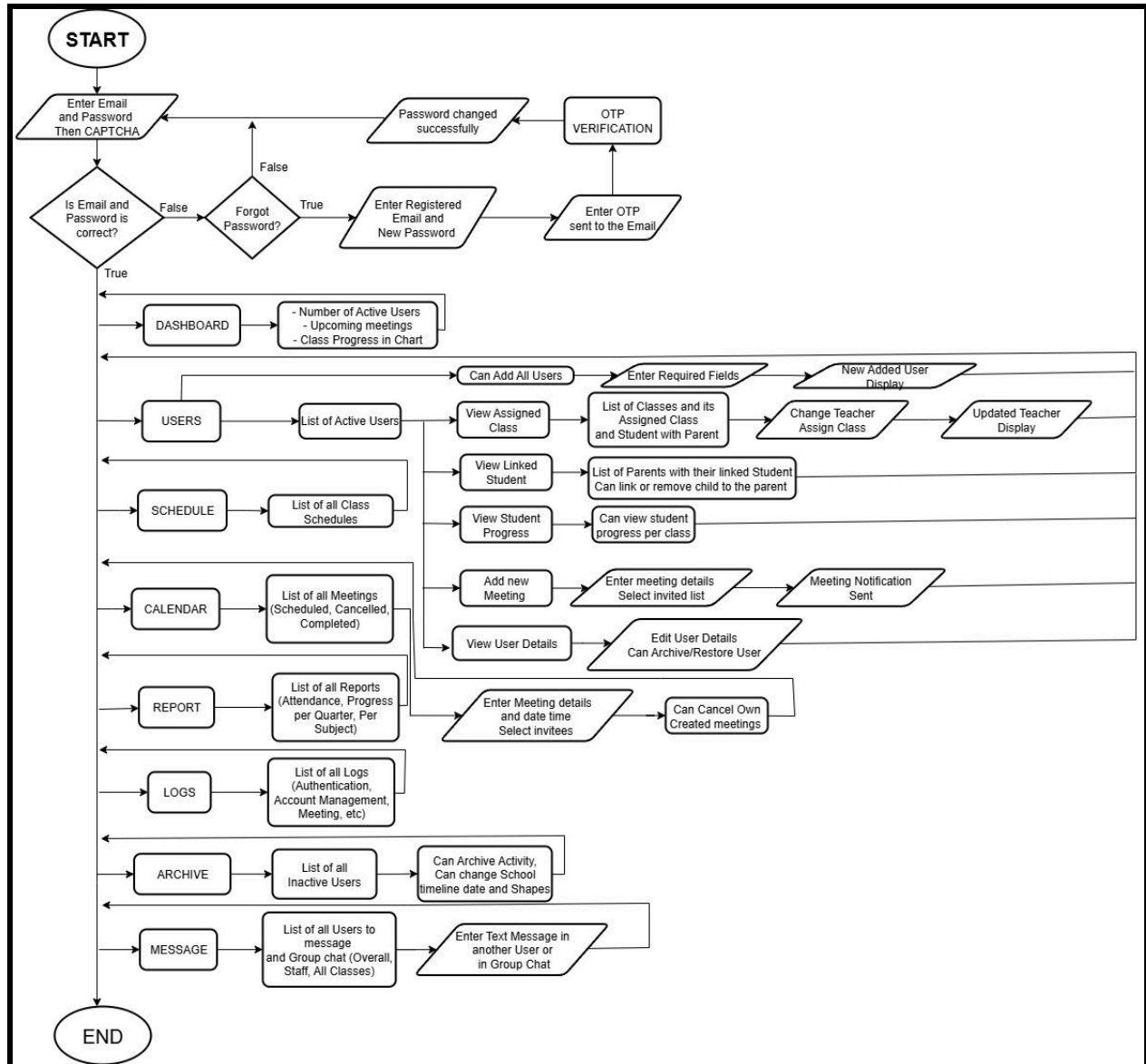
Proposed System Flowchart





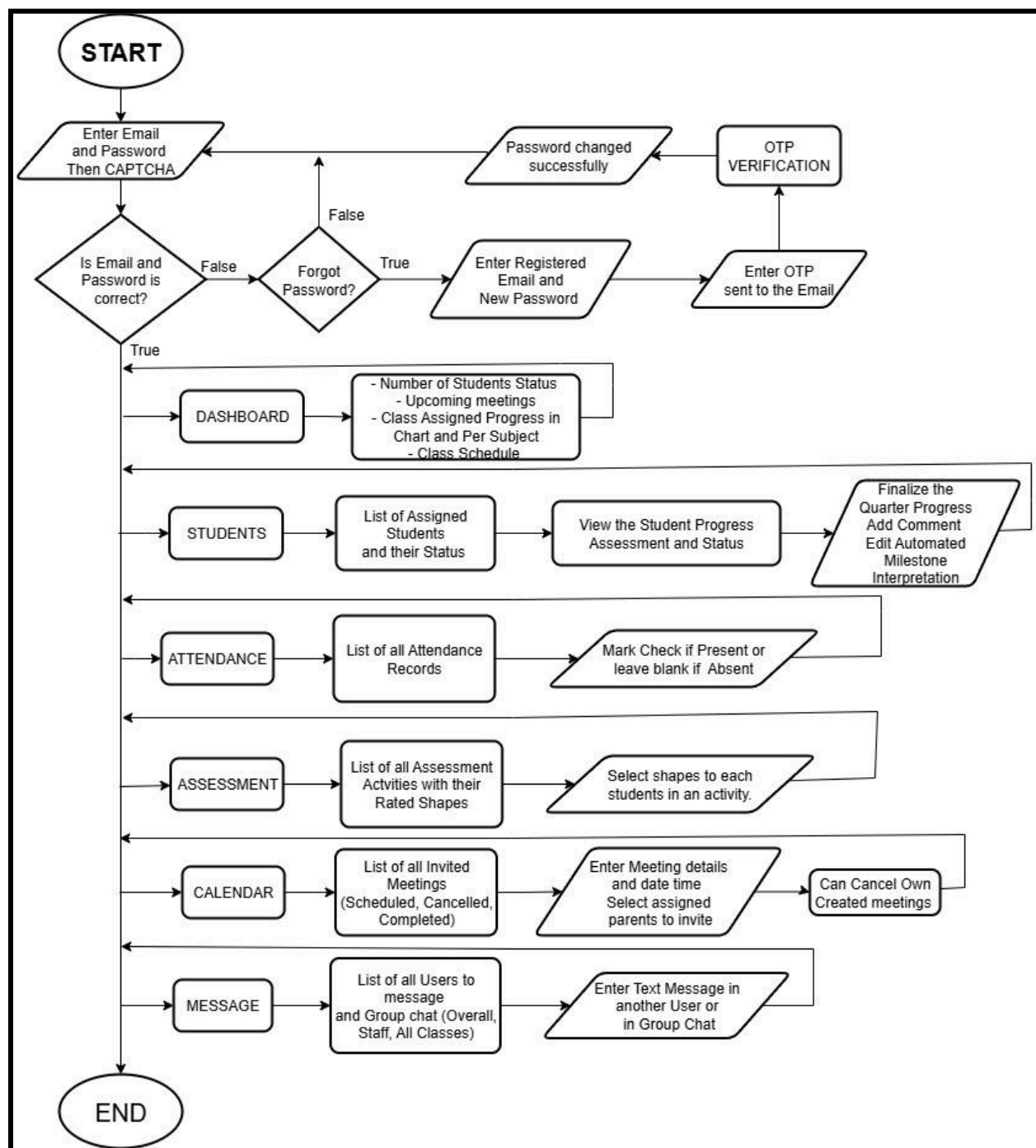
Appendix L

System Activity Diagram Admin



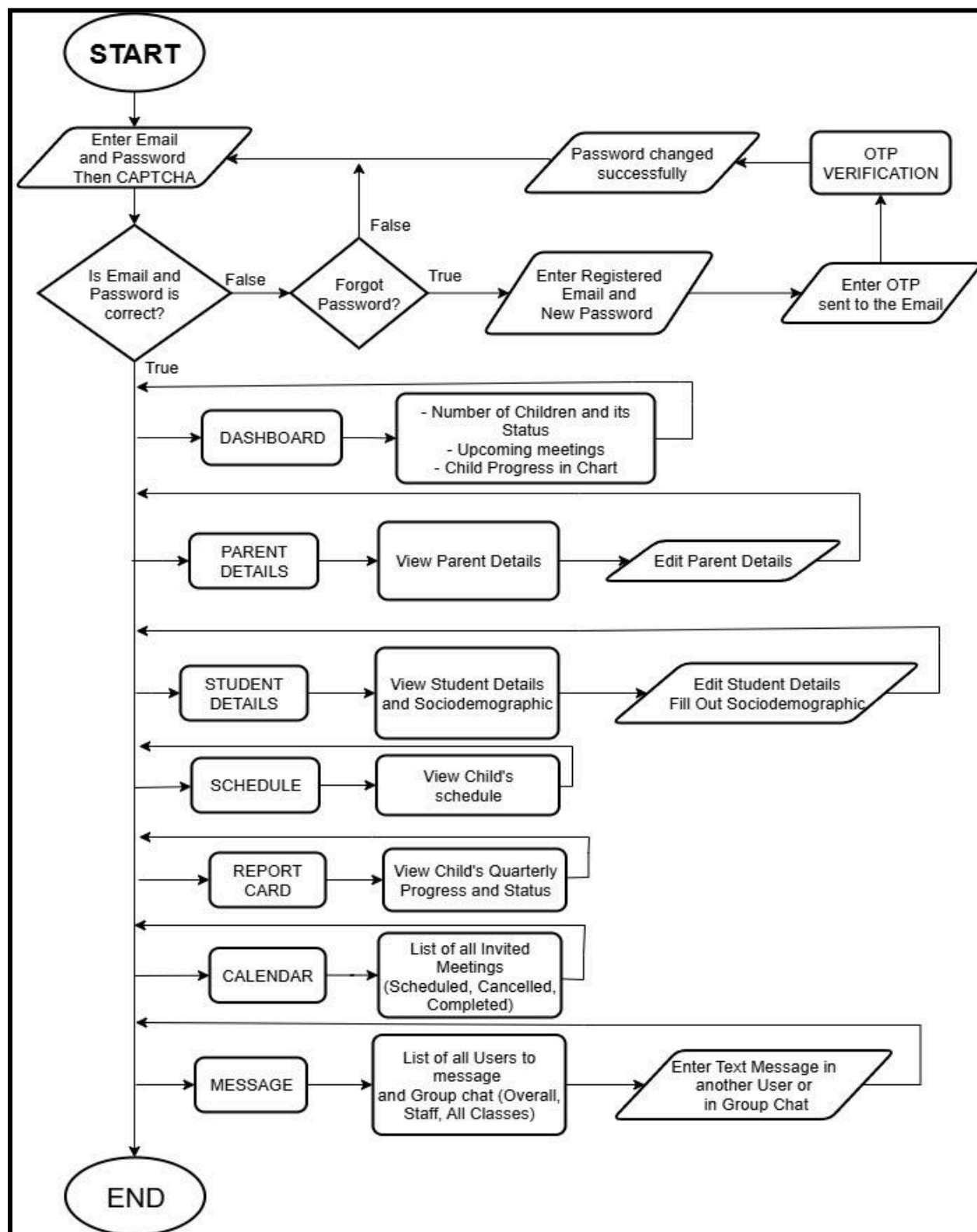


Teacher



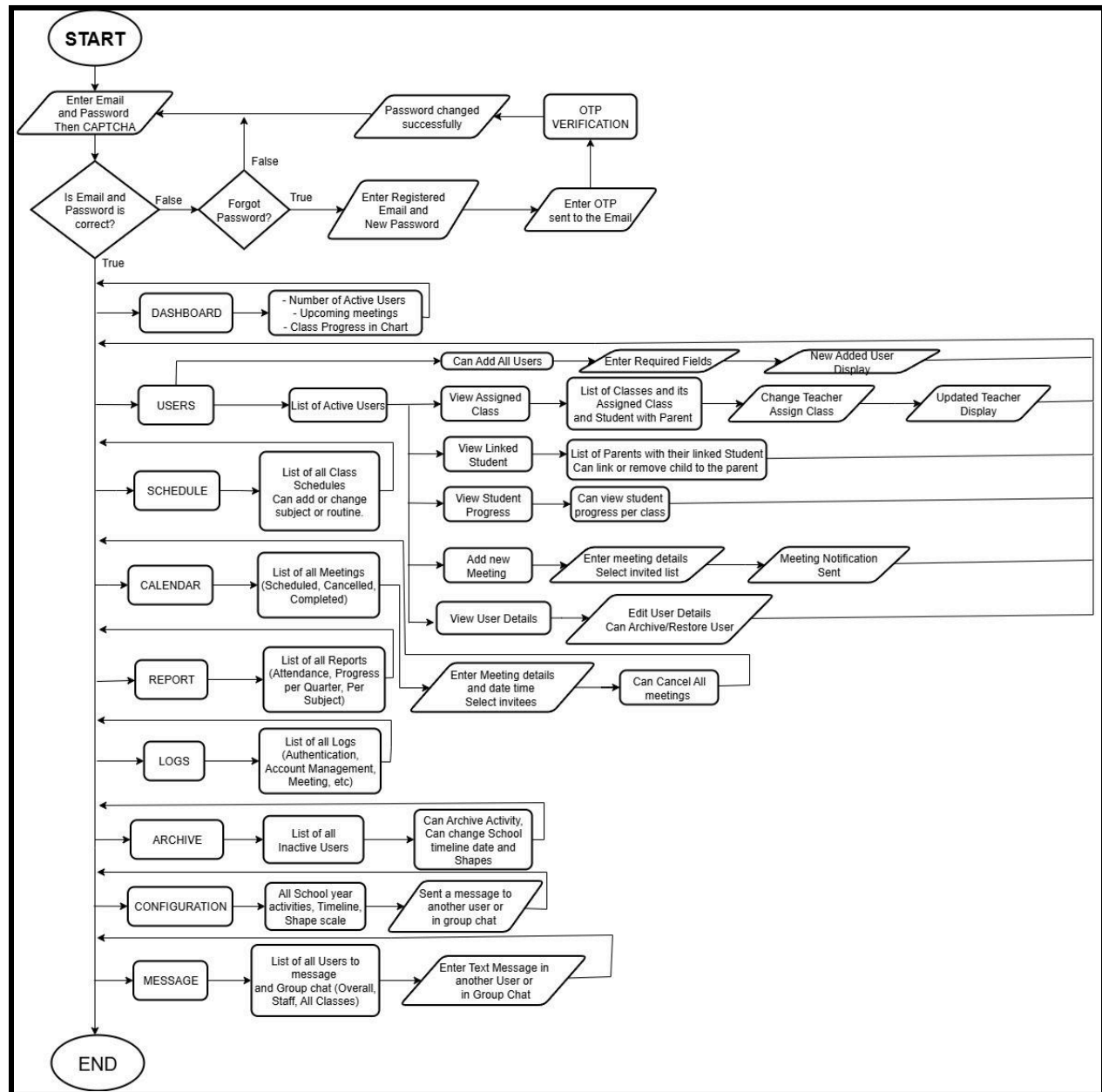


Parent





Super Admin





Appendix M

Data Dictionary

Field Name	Type	Length	Description
user_id	INT	11	Unique ID of the user (Primary Key, Auto Increment).
user_role	INT	11	Role ID (links to tbl_roles.role_id) to define if user is Super Admin, Admin, Teacher, or Parent.
user_pass	VARCHAR	255	Encrypted user password for system login.
user_firstname	VARCHAR	100	User's first name.
user_middlename	VARCHAR	100	User's middle name
user_lastname	VARCHAR	100	User's last name
user_birthdate	DATE	—	User's date of birth.
user_email	VARCHAR	150	User's email address (used for login).
user_contact_no	VARCHAR	20	User's contact number
user_photo	TEXT	—	Profile picture of the user
user_status	ENUM	('Active','Inactive')	Indicates if the user account is active or inactive.
is_new	ENUM	('Yes','No')	Indicates if the account is newly



			created (e.g., first-time login).
created_at	TIMESTAMP	—	Date and time when the account was created (default: current timestamp).

Field Name	Type	Length	Description
role_id	INT	11	Unique identifier for each role
role_name	VARCHAR	50	Role name (e.g., Super Admin, Admin, Teacher, Parent)

Field Name	Type	Length	Description
student_id	INT	11	Unique identifier for each student.
parent_id	INT	11	References tbl_users.user_id to link the student's account to their parent/guardian.
parent_profile_id	INT	11	tbl_parents_profile.parent_profile_id for parent profile details.
level_id	INT	11	References



			tbl_student_levels.level_id to indicate the student's academic level/grade.
stud_firstname	VARCHAR	100	First name of the student.
stud_middlename	VARCHAR	100	Middle name of the student.
stud_lastname	VARCHAR	100	Last name of the student.
stud_birthdate	DATE	—	Date of birth of the student.
stud_enrollment_date	DATE	—	Date when the student was officially enrolled.
stud_handedness	ENUM	—	Student's dominant hand.
stud_gender	ENUM	—	Gender of the student (Male, Female).
stud_schedule_class	ENUM	—	Student's class schedule (Morning, Afternoon).
stud_photo	TEXT	—	Path or file reference to the student's profile photo.
stud_school_status	ENUM	—	Current school status of the student (Active, Inactive). Default is Active.



Field Name	Type	Length	Description
activity_id	INT	11	Primary key - activity identifier
advisory_id	INT	11	Foreign key - tbl_advisory.advisory_id
quarter_id	INT	11	Foreign key - tbl_quarters.quarter_id
subject_id	INT	11	Foreign key - tbl_subjects.subject_id
activity_num	INT	11	Activity sequence/number
activity_name	VARCHAR	100	Activity title/name
activity_date	DATE	—	Schedule activity date
activity_status	VARCHAR	100	Status (example default: Active)

Field Name	Type	Length	Description
quarter_id	INT	11	Primary key — quarter identifier
quarter_name	VARCHAR	100	Quarter label/name (e.g., Q1)
start_date	DATE	—	Quarter start date
end_date	DATE	—	Quarter end date



Field Name	Type	Length	Description
subject_id	INT	11	Primary key — subject identifier
subject_name	VARCHAR	100	Subject name (e.g., Early Math, Language)

Field Name	Type	Length	Description
advisory_id	INT	11	Primary key — advisory/class identifier
level_id	INT	11	Foreign key - tbl_student_levels.level_id
lead_teacher_id	INT	11	Foreign key - tbl_users.user_id (lead teacher)
assistant_teacher_id	INT	11	Foreign key - tbl_users.user_id (assistant teacher)
total_male	INT	11	Count of male students (default 0)
total_female	INT	11	Count of female students (default 0)
total_students	INT	11	Total students (stored/generated)

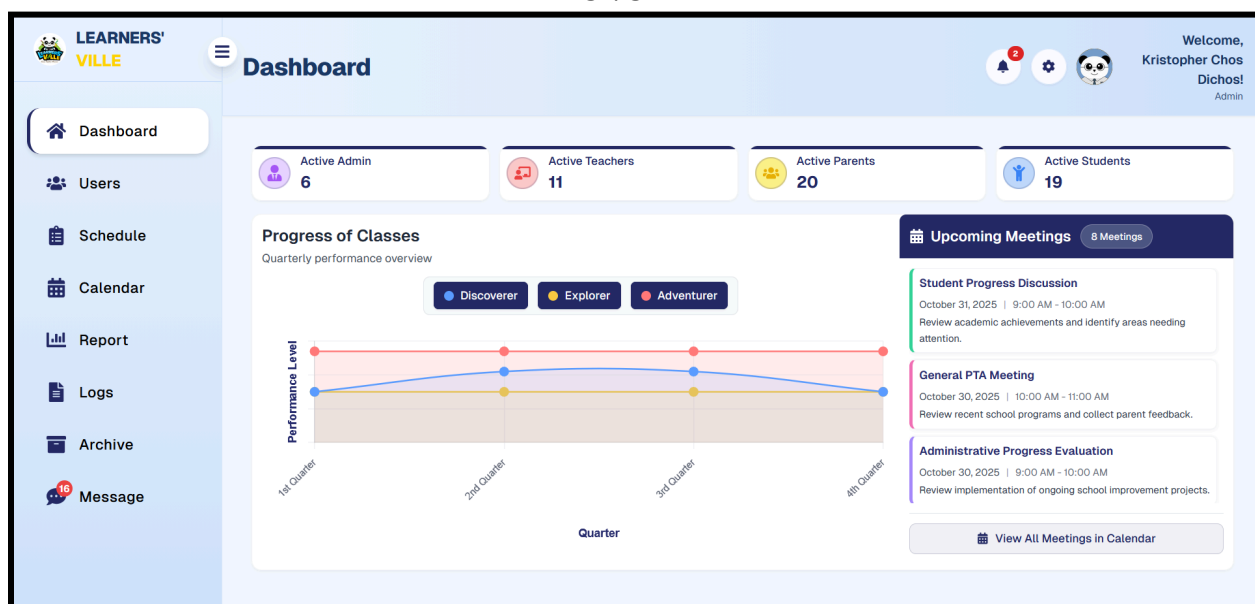


Field Name	Type	Length	Description
add_info_id	INT	11	Primary key — additional info record id
user_id	INT	11	Foreign key - tbl_users.user_id
barangay	VARCHAR	100	Barangay value (address detail)
city_municipality	VARCHAR	100	City/municipality
province	VARCHAR	100	Province
country	VARCHAR	100	Country
tin_number	VARCHAR	50	TIN number
sss_number	VARCHAR	50	SSS number
pagibig_number	VARCHAR	50	Pag-IBIG number



Appendix N

UI/UX



LEARNERS' VILLE Users

Welcome, Kristopher Chos Dichos! Admin

Search by name... 11 Teachers [Add User](#)

Teacher Parent Student Table Cards [View Assigned Class](#)

Photo	Full Name	Email	Assigned Class
	Borres, Angela Teresa	angelaborres@gmail.com	Non Advisory
	Caamino, Janneth	jannethcaamino@gmail.com	Adventurer
	Decena, Jessa Hambora	decenajessa03@gmail.com	Explorer
	Ebarat, Johanna Puerto	johannaebarat@gmail.com	Explorer
	Enfeda, Delfin	v.vrind@gmail.com	Non Advisory



Schedule

Welcome, Kristopher Chos Dichos! Admin

Dashboard

Users

Schedule

Calendar

Report

Logs

Archive

Message

Class Schedule of Discoverer

Select Class

Time	Monday 10:30 AM - 11:30 PM/1:30 PM - 2:30 PM	Wednesday 10:30 AM - 11:30 PM/1:30 PM - 2:30 PM	Friday 10:30 AM - 11:30 PM/1:30 PM - 2:30 PM
• 15 minutes	Attendance / Circle time	Attendance / Circle time	Attendance / Circle time
• 15 minutes	Early Math / Socio Emotional	Early Math / Socio Emotional	Science
• 10 minutes	Snack time	Snack time	Snack time
• 15 minutes	Literacy	Literacy	Basic Life Skills / Physical Activities
• 3 minutes	Story time / Bubble time	Story time / Bubble time	Story time / Bubble time
• 2 minutes	Wrapping up / Goodbye	Wrapping up / Goodbye	Wrapping up / Goodbye

Calendar

Welcome, Kristopher Chos Dichos! Admin

Dashboard

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Today Back Next

October 2025

Month Week Day Agenda

Mon	Tue	Wed	Thu	Fri	Sat	Sun
29	30	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2

Schedule Details

Scheduled

Student Progress Discussion

October 31, 2025 | 09:00 AM - 10:00 AM

Scheduled

General PTA Meeting

October 30, 2025 | 10:00 AM - 11:00 AM

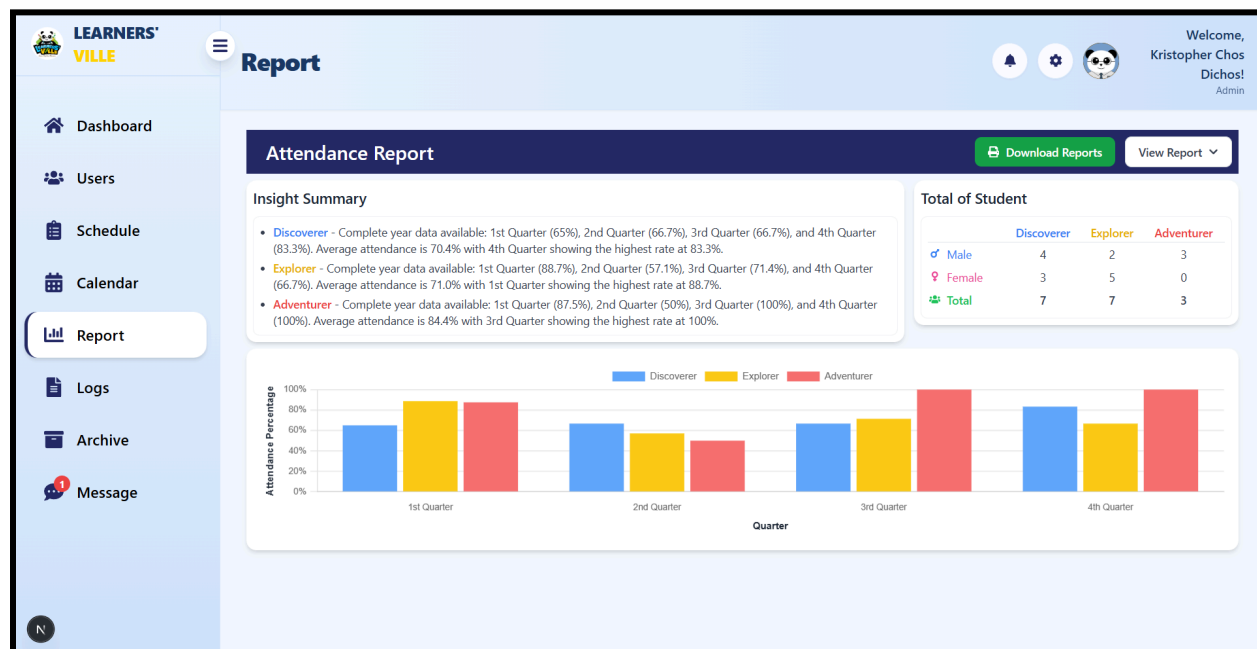
Rescheduled

Administrative Progress Evaluati...

October 30, 2025 | 09:00 AM - 10:00 A...

Scheduled

Anarlemic Drinnace Consultation





System Logs

Welcome,
Kristopher Chos Dichos!
Admin

Dashboard

Users

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Calendar

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Logs

Archive

Message

Timestamp

Description

User

Entity

10/4/2025, 5:07:43 PM

Kristopher Chos Dichos login to the system

Kristopher Chos Dichos

Authentication

10/4/2025, 3:40:42 PM

Kristopher Chos Dichos login to the system

Kristopher Chos Dichos

Authentication

10/4/2025, 3:30:21 PM

James Anthony Placido login to the system

James Anthony Placido

Authentication

10/4/2025, 2:14:12 PM

Kristopher Chos Dichos login to the system

Kristopher Chos Dichos

Authentication

10/4/2025, 2:13:48 PM

Changed password

Kristopher Chos Dichos

Security

10/4/2025, 2:08:24 PM

Jessa Hambora Decena login to the system

Jessa Hambora Decena

Authentication

10/4/2025, 1:35:19 AM

Andrew Lara Nerona login to the system

Andrew Lara Nerona

Authentication

Showing 1-7 of 441

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Archive

Welcome,
Kristopher Chos Dichos!
Admin

Dashboard

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Message

Staff

Parent

Student

Search by name, email, or contact number

Total Archived: 6

Archived Staff (Administrators and Teachers)

Staff: 4

Full Name

User Type

Email

Contact Number

Action

Johans, Chris

Admin

johans@gmail.com

Not specified

Roms, Juliet

Admin

juliet@gmail.com

+63 928 349 4757

Torres, Grace Fernandez

Teacher

grace@gmail.com

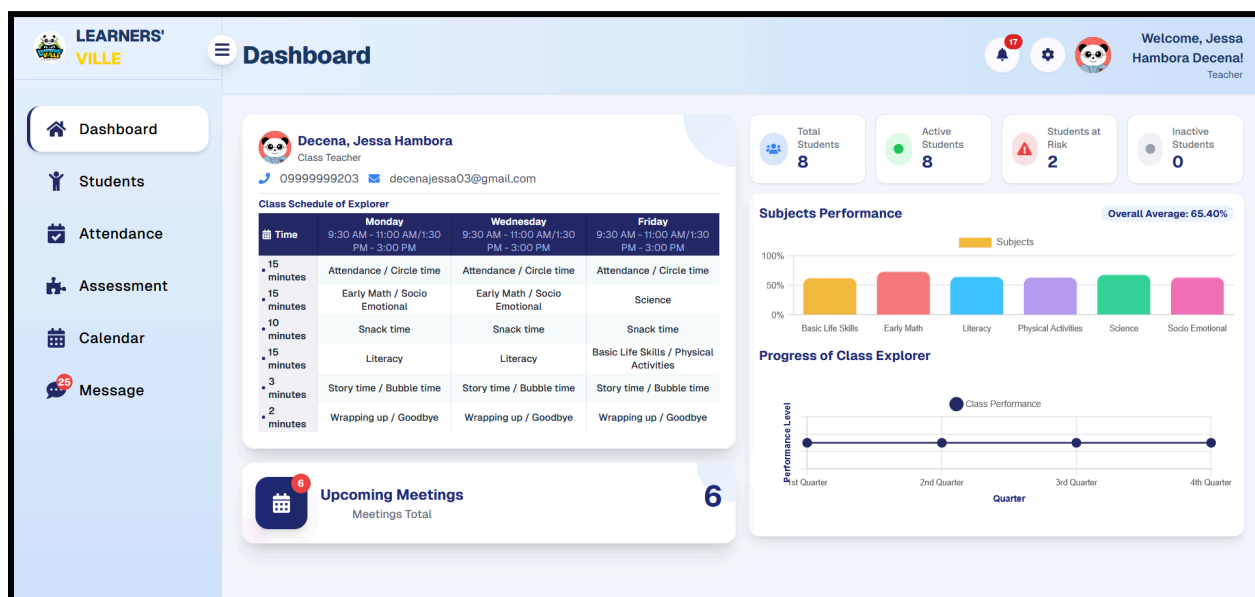
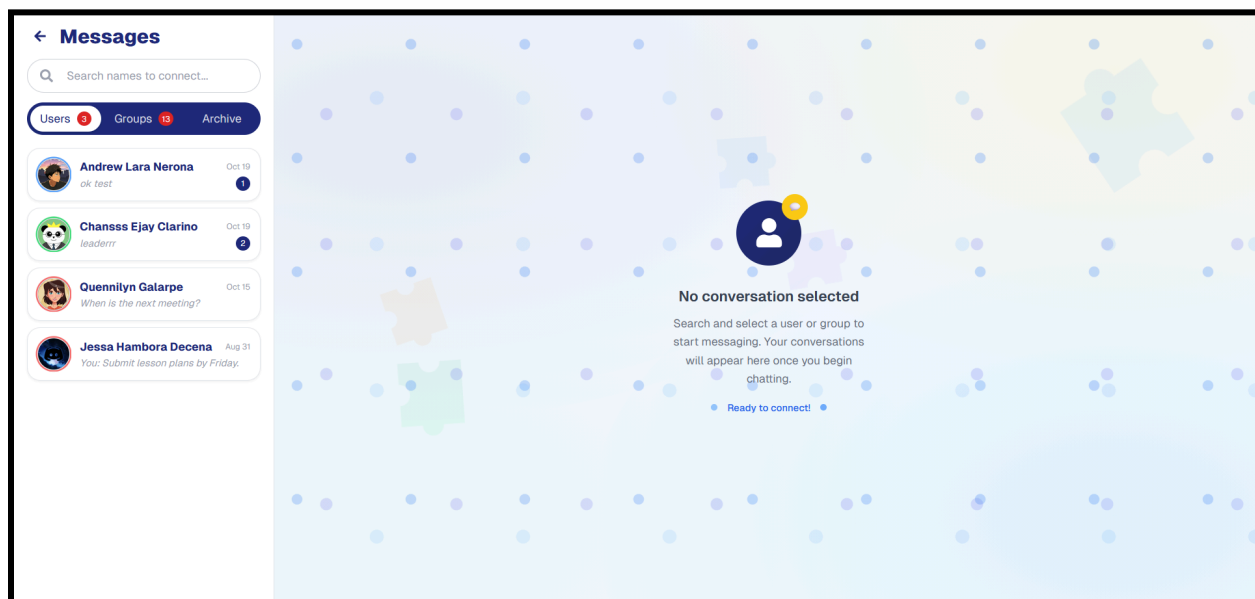
+63 937 123 4567

Turtoza, Fritz

Admin

fritz@gmail.com

Not specified





Students

Welcome, Jessa
Hambora Decena!
Teacher

Dashboard

Students

Attendance

Assessment

Calendar

Message

Search Students

Session

All

Lead Teacher: Hambora, Decena, Jessa

Assistant Teacher: Puerto, Ebarat, Johanna

Explorer

Male: 2
Female: 6
Total: 8

Student Name	Gender	Session	Risk Level
Bacsarsa, Tine Grace Israel	Female	Morning	Low
Bali, Junabel Paca	Male	Morning	Moderate
Bogarts, Angely Eve	Female	Afternoon	No Data
Gamon, Richard Montizor	Male	Morning	Low
Lopez, Zoe Jimenez	Female	Afternoon	Low
Madrenos, Gleiah Jimenez	Female	Morning	Low

Attendance

Welcome, Jessa
Hambora Decena!
Teacher

Dashboard

Students

Attendance

Assessment

Calendar

Message

Session

Morning

5 Students

Mark Attendance

Student Name	Feb 12 2026	Feb 11 2026	Feb 10 2026	Dec 11 2025	Dec 10 2025
Bacsarsa, Tine Grace Israel	Present	Present	Absent	Absent	Absent
Bali, Junabel Paca	Absent	Absent	Absent	Present	Present
Gamon, Richard Montizor	Absent	Present	Present	Present	Present
Madrenos, Gleiah Jimenez	Absent	Present	Present	Present	Present
Placido, Jamica Mica	Present	Present	Absent	Present	Present

<

1

2

3

4

>



Assessment

Welcome, **Jessa Hambora Decena!**
Teacher

Dashboard

Students

Attendance

Assessment

Calendar

Message

Subject

Session

Quarter

5 Students

Add Activity

Student Name	Activity 13 Sep 14, 2025	Activity 12 Aug 21, 2025	Activity 11 Aug 20, 2025	Activity 10 Aug 18, 2025	Activity 9 Aug 17, 2025	Activity 8 Aug 16, 2025	Activity 7 Aug 15, 2025	Activity 6 Aug 14, 2025	Activity 5 Aug 13, 2025	Activity 4 Aug 12, 2025
Bacsarsa, Tine Grace Israel	🔴	🔴	🟢	🟢	🟡	🟢	🟡	🟢	🟡	🟡
Bali, Junabel Paca	🔴	🟡	🟡	🟢	🔵	🟡	🟡	🟡	🔴	🟢
Gamon, Richard Montizor	🟡	🟡	🟡	🔴	🔴	🔴	🔴	🔴	🔴	🔴
Madrenos, Gleiah Jimenez	🔴	🟡	🔴	🟡	🔴	🔴	🔴	🔴	🟡	🟡
Placido, Jamica Mica	🟡	🟡	🟡	🟢	🟢	🟡	🟡	🟢	🔵	🔴

<

1

2

>

Calendar

Welcome, **Jessa Hambora Decena!**
Teacher

Dashboard

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Attendance

Assessment

Calendar

Message

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October 2025

Month

Week

Day

Agenda

Mon	Tue	Wed	Thu	Fri	Sat	Sun
29	30	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2

Schedule Details

Scheduled

Student Progress Discussi...

October 31, 2025 | 09:00 AM - 10:00 AM

Scheduled

General PTA Meeting

October 30, 2025 | 10:00 AM - 11:00 AM

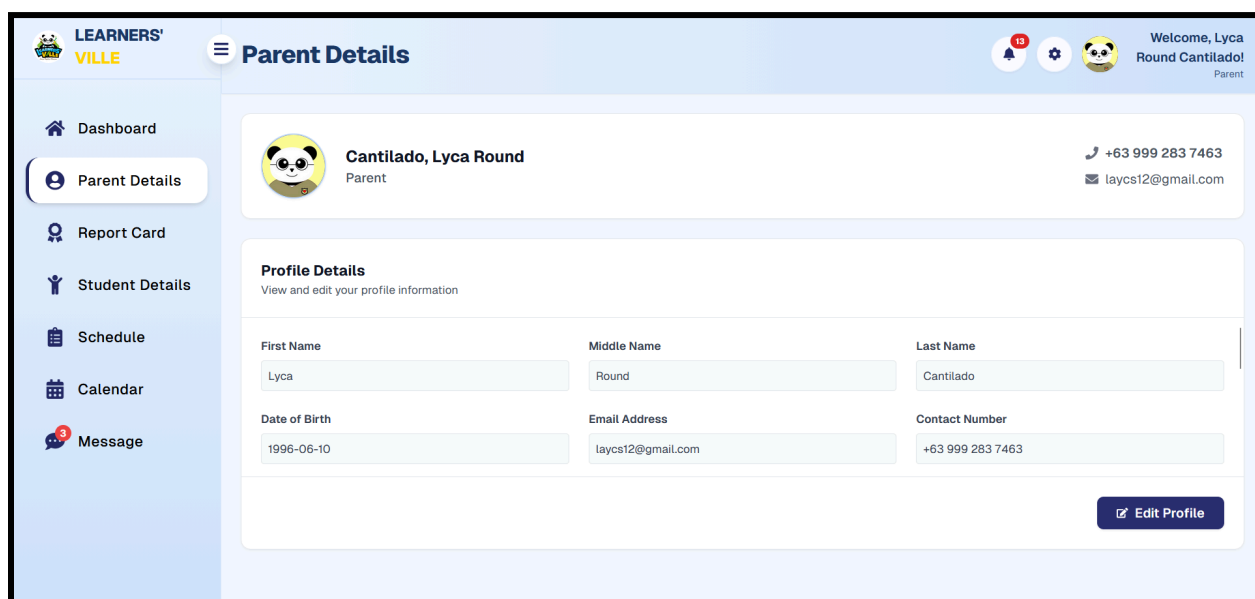
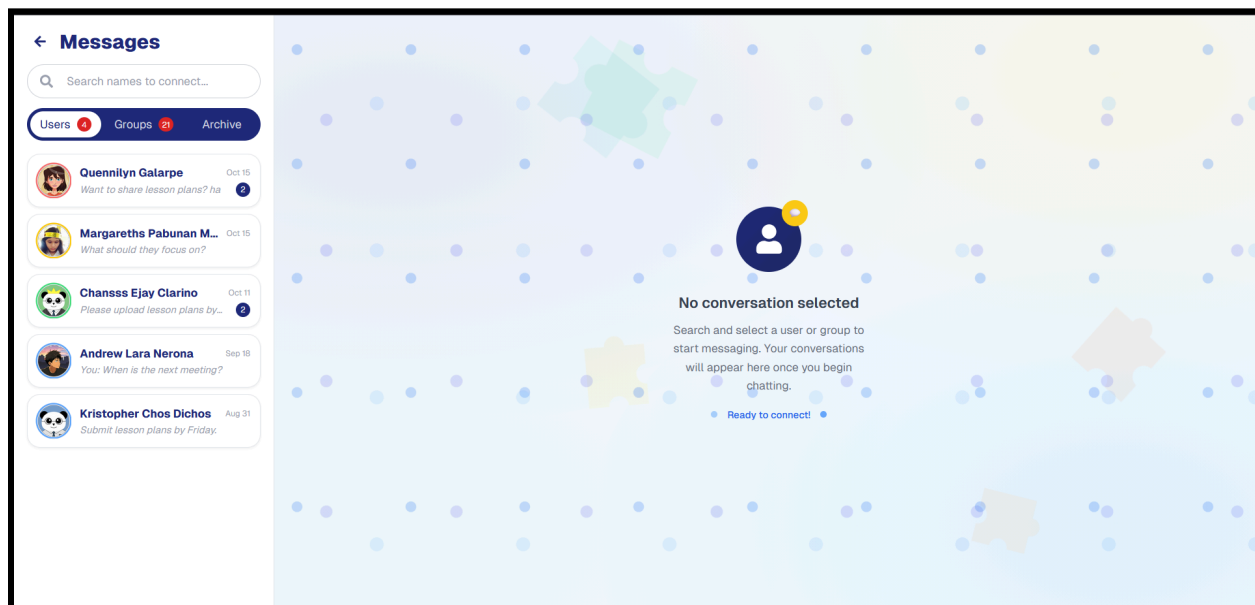
Rescheduled

Administrative Progress Evaluati...

October 30, 2025 | 09:00 AM - 10:00 A...

Scheduled

Academic Progress Consultation





Report Card

Welcome, Lyca Round Cantilado!
Parent

Dashboard

Parent Details

Report Card

Student Details

Schedule

Calendar

Message

Gamon, Richard Montizor

Schedule: Morning Gender: Male Handedness: Right Date of Birth: July 2, 2022
Lead Teacher: Hambora, Decena, Jessa Assistant Teacher: Puerto, Ebarat, Johanna

Assessment

Status

Download PDF

Quarterly Assessment
Subject performance across quarters

Subjects	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	Final
Basic Life Skills	★	★	♥	♥	♥
Early Math	♥	♥	♥	♥	♥
Literacy	★	◆	★	♥	★
Physical Activities	★	★	♥	♥	♥
Science	♥	♥	★	♥	♥
Socio Emotional	★	♥	♥	★	♥
Quarter Result	● ★	● ★	● ♥	● ♥	● ♥

Assessment Legend
Performance indicators and meanings

Shapes	Descriptions	Remarks
♥	- Excellent	- Passed
★	- Very Good	- Passed
◆	- Good	- Passed
▲	- Need Help	- Passed
●	- Not Met	- Failed

Student Details

Welcome, Lyca Round Cantilado!
Parent

Dashboard

Parent Details

Report Card

Student Details

Schedule

Calendar

Message

Sociodemographic Profile
View and edit student profile information

Basic Information

First Name

Middle Name

Last Name

Richard

Montizor

Gamon

Gender

Date of Birth

Enrollment Date

☒ Male
☐ Female

07 / 02 / 2022

06 / 29 / 2025

Session

Handedness

Photo

Morning

☒ Right
☐ Left

Edit Profile



Schedule

Welcome, Lyca
Round Cantilado!
Parent

Dashboard
 Parent Details
 Report Card
 Student Details
 Schedule
 Calendar
 Message

Class Schedule of Explorer

Time	Monday 9:30 AM - 11:00 AM	Wednesday 9:30 AM - 11:00 AM	Friday 9:30 AM - 11:00 AM
15 minutes	Attendance / Circle time	Attendance / Circle time	Attendance / Circle time
15 minutes	Early Math / Socio Emotional	Early Math / Socio Emotional	Science
10 minutes	Snack time	Snack time	Snack time
15 minutes	Literacy	Literacy	Basic Life Skills / Physical Activities
3 minutes	Story time / Bubble time	Story time / Bubble time	Story time / Bubble time
2 minutes	Wrapping up / Goodbye	Wrapping up / Goodbye	Wrapping up / Goodbye

Calendar

Welcome, Lyca
Round Cantilado!
Parent

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 Parent Details
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Today Back Next
 October 2025
 Month Week Day Agenda

Mon	Tue	Wed	Thu	Fri	Sat	Sun
29	30	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2

Schedule Details

Scheduled

Administrative Progress Evaluati...

October 30, 2025 | 09:00 AM - 10:00 AM

Scheduled

Academic Progress Consultation

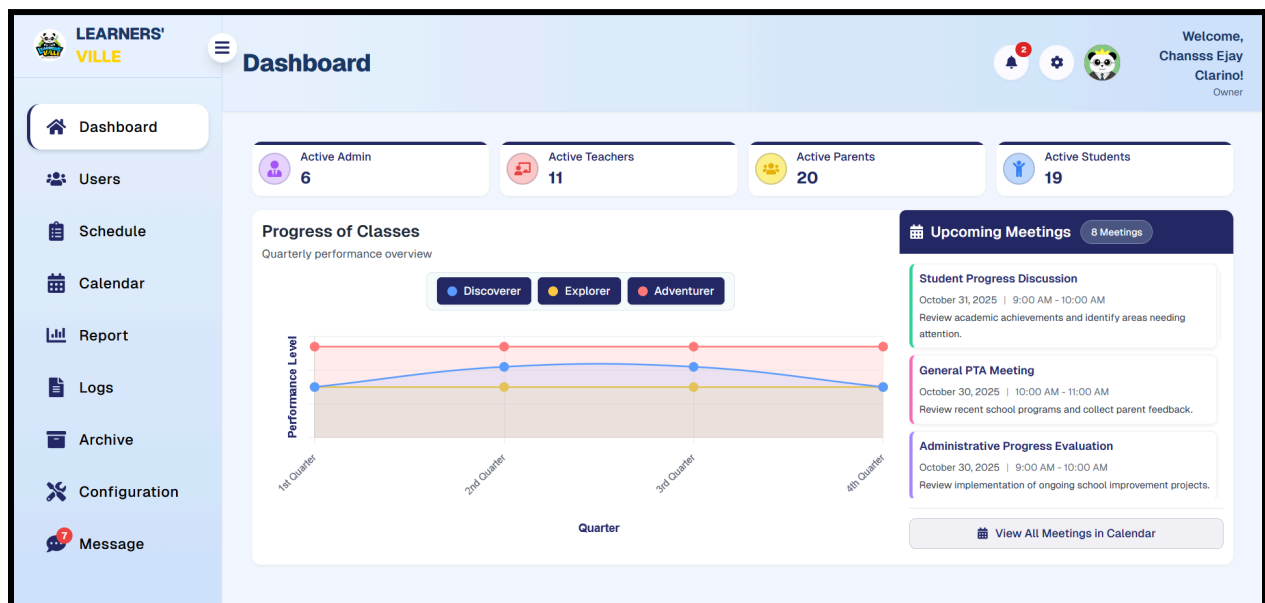
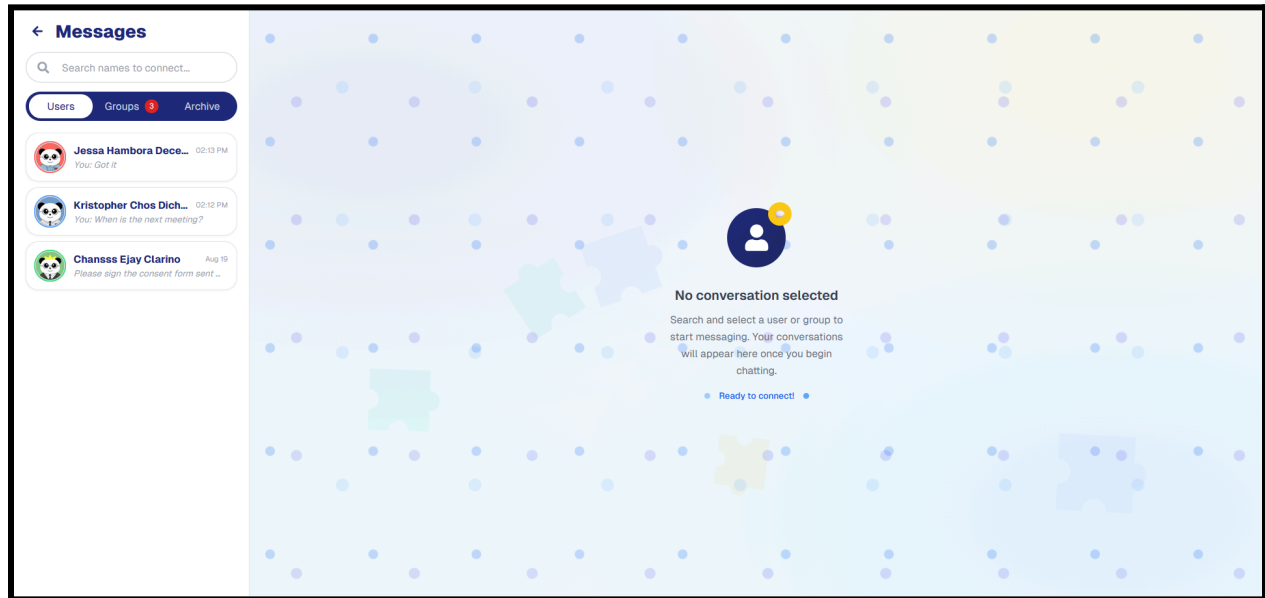
October 28, 2025 | 09:00 AM - 10:00 AM

Rescheduled

Class Officers' Assembly

October 23, 2025 | 09:00 AM - 10:00 AM

Scheduled





Users

Search by name...

6 Admins

Add User

Admin

Teacher

Parent

Student

Table

Cards

Photo	Full Name	Email	Contact Number
	Altahan, Mike	althan@gmail.com	09287367464
	Amorin, Elmer	amorin@gmail.com	09476736363
	Anches, Laurenz Montizor	laurenz@gmail.com	Not specified
	Baterna, Junel	junel@gmail.com	09293234632
	Dichos, Kristopher Chos	dichoskristopher@gmail.com	09918123333

Dashboard

Users

Schedule

Calendar

Report

Logs

Archive

Configuration

Message

Welcome, Chansss Ejay Clarino!

Owner

Schedule

+ Add Schedule Item

Select Class

Time	Monday 10:30 AM - 11:30 PM / 1:30 PM - 2:30 PM	Wednesday 10:30 AM - 11:30 PM / 1:30 PM - 2:30 PM	Friday 10:30 AM - 11:30 PM / 1:30 PM - 2:30 PM
• 15 minutes	Attendance / Circle time	Attendance / Circle time	Attendance / Circle time
• 15 minutes	Early Math / Socio Emotional	Early Math / Socio Emotional	Science
• 10 minutes	Snack time	Snack time	Snack time
• 15 minutes	Literacy	Literacy	Basic Life Skills / Physical Activities
• 3 minutes	Story time / Bubble time	Story time / Bubble time	Story time / Bubble time
• 2 minutes	Wrapping up / Goodbye	Wrapping up / Goodbye	Wrapping up / Goodbye

Dashboard

Users

Schedule

Calendar

Report

Logs

Archive

Configuration

Message

Welcome, Chansss Ejay Clarino!

Owner

74



Calendar

Dashboard

Users

Schedule

Calendar

Report

Logs

Archive

Configuration

Message

Today

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October 2025

Month

Week

Day

Agenda

Mon	Tue	Wed	Thu	Fri	Sat	Sun
29	30	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2

Schedule Details

Scheduled

Student Progress Discussion

October 31, 2025 | 09:00 AM - 10:00 AM

Scheduled

General PTA Meeting

October 30, 2025 | 10:00 AM - 11:00 AM

Rescheduled

Administrative Progress Evaluati...

October 30, 2025 | 09:00 AM - 10:00 A...

Scheduled

Report

Dashboard

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Attendance Report

Download Reports

View Report

Insight Summary

- Discoverer** - Complete year data available: 1st Quarter (65%), 2nd Quarter (66.7%), 3rd Quarter (66.7%), and 4th Quarter (83.3%). Average attendance is 70.4% with 4th Quarter showing the highest rate at 83.3%.
- Explorer** - Complete year data available: 1st Quarter (88.7%), 2nd Quarter (55.6%), 3rd Quarter (71.4%), and 4th Quarter (66.7%). Average attendance is 70.6% with 1st Quarter showing the highest rate at 88.7%.
- Adventurer** - Complete year data available: 1st Quarter (87.5%), 2nd Quarter (50%), 3rd Quarter (100%), and 4th Quarter (100%). Average attendance is 84.4% with 3rd Quarter showing the highest rate at 100%.

Total of Student

	Discoverer	Explorer	Adventurer
♂ Male	4	2	3
♀ Female	3	6	1
👤 Total	7	8	4

Attendance Percentage

Discoverer Explorer Adventurer

3rd Quarter Discoverer: 66.7

Quarter	Discoverer	Explorer	Adventurer
1st Quarter	65%	88.7%	87.5%
2nd Quarter	66.7%	55.6%	50%
3rd Quarter	66.7%	71.4%	100%
4th Quarter	83.3%	66.7%	100%



System Logs

Welcome,
Chansss Ejay
Clarino!
Owner

Dashboard

Users

Schedule

Calendar

Report

Logs

Archive

Configuration

Message

Filter
All

Timestamp	Description	User	Entity
10/21/2025, 10:42:03 AM	Chansss Ejay Clarino edited the details of th...	Chansss Ejay Clarino	Account Management
10/21/2025, 10:25:45 AM	Chansss Ejay Clarino login to the system	Chansss Ejay Clarino	Authentication
10/20/2025, 9:52:47 AM	Margareths Pabunan Manongdo edited their ...	Margareths Pabunan Manong...	Account Management
10/19/2025, 1:59:41 PM	Andrew Lara Nerona logout from the system	Andrew Lara Nerona	Authentication
10/19/2025, 1:59:37 PM	Chansss Ejay Clarino logout from the system	Chansss Ejay Clarino	Authentication
10/19/2025, 12:21:36 PM	Chansss Ejay Clarino login to the system	Chansss Ejay Clarino	Authentication
10/19/2025, 12:21:31 PM	Kristopher Chos Dichos logout from the syst...	Kristopher Chos Dichos	Authentication

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Archive

Welcome,
Chansss Ejay
Clarino!
Owner

Dashboard

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Calendar

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Logs

Archive

Configuration

Message

Staff

Parent

Student

Total Archived: 6

Archived Staff (Administrators and Teachers)

Staff: 4

Full Name	User Type	Email	Contact Number	Action
Johans, Chris	Admin	johans@gmail.com	Not specified	
Roms, Juliet	Admin	juliet@gmail.com	+63 928 349 4757	
Torres, Grace Fernandez	Teacher	grace@gmail.com	+63 937 123 4567	
Turtoza, Fritz	Admin	fritz@gmail.com	Not specified	



System Configuration

Activities

School Year

Visual Scoring

System Configuration

All Activities

Total: 186

Active: 186

Archived: 0

Bulk Archive

Activity Name	Subject	Date	Quarter	Advisory Class	Status
March, Skip, and Hop Practice	Physical Activities	Mar 22, 2026	4th Quarter	Explorer	Active
Jump Over Lines or Ropes	Physical Activities	Mar 20, 2026	4th Quarter	Explorer	Active
My Favorite Things Sharing Time	Socio Emotional	Mar 07, 2026	4th Quarter	Explorer	Active
Emotion Wheel	Socio Emotional	Mar 06, 2026	4th Quarter	Explorer	Active
Water the Plants	Basic Life Skills	Mar 02, 2026	4th Quarter	Explorer	Active

Dashboard

Users

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Configuration

Message

Welcome, Chansss Ejay Clarino!

Owner

Messages

Users

Groups

Archive

Kristopher Chos Dichos

Oct 19

You: leaderrr

Andrew Lara Nerona

Oct 15

You: Please coordinate with teache...

Quennilyn Galarpe

Oct 15

Any development programs?

Margareths Pabunan M...

Oct 15

You: Thank you for reaching out.

Jessa Hambora Decena

Oct 11

You: Please upload lesson plans b...

Clyde Denzel Parol

Sep 18

Concerned about curriculum.

Johanna Puerto Ebarat

Aug 31

Explain new policies?

Lyca Round Cantilado

Aug 19

No conversation selected

Search and select a user or group to start messaging. Your conversations will appear here once you begin chatting.

Ready to connect!



Appendix O

Code Snippets

```
try {
    $conn->beginTransaction();

    // Map role to numeric user_role
    $roleMap = ['admin' => 2, 'teacher' => 3, 'parent' => 4];
    $userRole = $roleMap[$role] ?? null;

    if (!$userRole) {
        throw new Exception('Invalid role specified.');
```

```
    }
```

```
    // Insert into tbl_users
```

```
    $stmt = $conn->prepare("INSERT INTO tbl_users (
        user_role, user_pass, user_firstname, user_middlename, user_lastname, user_birthdate,
        user_email, user_contact_no, user_photo, user_status, is_new
    ) VALUES (
        ?, ?, ?, ?, ?, ?, ?, ?, ?, 'Active', 'Yes'
    )");
```

```
    $stmt->execute([
        $userRole,
        $hashedPassword,
        $firstname,
        $middlename,
        $lastname,
        $birthdate,
        $email,
        $contact,
        $photo
```

```
    ]);
```

```
$user_id = $conn->lastInsertId();
```

```
// Admin & Teacher: Insert into tbl_add_info with address + gov IDs
```

```
if (in_array($role, ['admin', 'teacher'])) {
    $stmt2 = $conn->prepare("INSERT INTO tbl_add_info (
        user_id, tin_number, sss_number, pagibig_number, barangay, city_municipality, province, country
    ) VALUES (?, ?, ?, ?, ?, ?, ?, ?)");
```

```
    $stmt2->execute([$user_id, $tin, $sss, $pagibig, $barangay, $city, $province, $country]);
```

```
}
```

```
// Parent: Insert into tbl_parents_profile with address only
```

```
if ($role === 'parent') {
    $stmt3 = $conn->prepare("INSERT INTO tbl_parents_profile (
        user_id, barangay, city_municipality, province, country
    ) VALUES (?, ?, ?, ?, ?)");
```

```
    $stmt3->execute([$user_id, $barangay, $city, $province, $country]);
```

```
}
```



```
// Begin transaction
$conn->beginTransaction();

// Insert into tbl_students with parent_id as NULL
$stmtStudent = $conn->prepare("INSERT INTO tbl_students (
    parent_id, level_id,
    stud_firstname, stud_middlename, stud_lastname, stud_birthdate,
    stud_enrollment_date, stud_handedness, stud_gender,
    stud_schedule_class, stud_photo, stud_school_status
) VALUES (NULL, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)");

$stmtStudent->execute([
    $levelId,
    $firstName,
    $middleName,
    $lastName,
    $birthDate,
    $enrollDate,
    $handedness,
    $gender,
    $scheduleClass,
    $photo,
    $schoolStatus
]);

$studentId = $conn->lastInsertId();
$conn->commit();
```



```
try {
    // Insert the system log record with proper target IDs
    $query = $conn->prepare("
        INSERT INTO tbl_system_logs (user_id, target_user_id, target_student_id, action, timestamp)
        VALUES (:user_id, :target_user_id, :target_student_id, :action, NOW())
    ");

    if ($userId === null) {
        $query->bindValue(":user_id", null, PDO::PARAM_NULL);
    } else {
        $query->bindValue(":user_id", $userId, PDO::PARAM_INT);
    }

    // Bind target IDs based on action type
    if ($targetUserId !== null) {
        $query->bindValue(":target_user_id", $targetUserId, PDO::PARAM_INT);
    } else {
        $query->bindValue(":target_user_id", null, PDO::PARAM_NULL);
    }

    if ($targetStudentId !== null) {
        $query->bindValue(":target_student_id", $targetStudentId, PDO::PARAM_INT);
    } else {
        $query->bindValue(":target_student_id", null, PDO::PARAM_NULL);
    }

    $query->bindValue(":action", $action, PDO::PARAM_STR);

    if ($query->execute()) {
        // Log successful operation for debugging
        error_log("System log created successfully: User ID: " . ($userId ?? 'NULL') . ", Action: $action, Target User ID: ");

        echo json_encode([
            "success" => true,
            "message" => "System log created successfully",
            "log_id" => $conn->lastInsertId()
        ]);
    }
}
```

Cursor Tab Paul Andrew Nerona (1 month ago) OVR Ln 38, C



```
try {
    $conn->beginTransaction();

    // 1. Insert into tbl_meetings
    $stmt = $conn->prepare("
        INSERT INTO tbl_meetings (
            meeting_title, parent_id, student_id, advisory_id,
            meeting_start, meeting_end, meeting_agenda, meeting_status
        ) VALUES (?, ?, ?, ?, ?, ?, ?, 'Scheduled')
    ");
    $stmt->execute([
        $title,
        $parent_id,
        $student_id,
        $advisory_id,
        $start,
        $end,
        $agenda
    ]);
    $meeting_id = $conn->lastInsertId();

    // 2. Insert into tbl_notifications
    $notif_message = $data['notif_message'] ?? "[MEETING] Created the Meeting";
    $stmt = $conn->prepare("
        INSERT INTO tbl_notifications (
            notif_message, created_by, created_at, meeting_id
        ) VALUES (?, ?, NOW(), ?)
    ");
    $stmt->execute([
        $notif_message,
        $created_by,
        $meeting_id
    ]);
    $notification_id = $conn->lastInsertId();
}
```



```

try {
    // 3. Insert into tbl_notification_recipients
    $stmt = $conn->prepare("
        INSERT INTO tbl_notification_recipients (notification_id, user_id, recipient_type)
        VALUES (?, ?, ?)
    ");
    if (!empty($recipients)) {
        // Group meeting: insert all recipients
        foreach ($recipients as $r) {
            $stmt->execute([
                $notification_id,
                $r['user_id'],
                $r['recipient_type']
            ]);
        }
    } else if ($parent_id && $advisory_id) {
        // 1-on-1: insert parent and both co-advisors (lead and assistant teacher) as recipients
        // Insert parent
        $stmt->execute([
            $notification_id,
            $parent_id,
            'Parent'
        ]);
        // Fetch co-advisors from tbl_advisory
        $advisoryStmt = $conn->prepare("SELECT lead_teacher_id, assistant_teacher_id FROM tbl_advisory WHERE advisory_id = ?");
        $advisoryStmt->execute([$advisory_id]);
        $advisory = $advisoryStmt->fetch(PDO::FETCH_ASSOC);
        if ($advisory) {
            if (!empty($advisory['lead_teacher_id'])) {
                $stmt->execute([
                    $notification_id,
                    $advisory['lead_teacher_id'],
                    'Teacher'
                ]);
            }
            if (!empty($advisory['assistant_teacher_id']) && $advisory['assistant_teacher_id'] != $advisory['lead_teacher_id']) {
                $stmt->execute([

```



```
try {
    // Fetch notifications from tbl_notifications
    $stmt = $conn->prepare("
        SELECT
            n.notification_id,
            n.notif_message,
            n.created_by,
            n.created_at,
            n.meeting_id,
            m.meeting_title,
            m.meeting_start,
            m.meeting_end,
            m.meeting_status,
            m.parent_id,
            m.student_id,
            m.advisory_id
        FROM tbl_notifications n
        LEFT JOIN tbl_meetings m ON n.meeting_id = m.meeting_id
        ORDER BY n.created_at DESC
        LIMIT 100
    ");
    $stmt->execute();
    $notifications = $stmt->fetchAll(PDO::FETCH_ASSOC);

    echo json_encode([
        'status' => 'success',
        'notifications' => $notifications
    ]);
} catch (PDOException $e) {
    http_response_code(500);
    echo json_encode([
        'status' => 'error',
        'message' => 'Database error',
        'error' => $e->getMessage()
    ]);
}
```



```
try {
    $notificationId = $input['notification_id'];
    $userRole = $input['user_role'] ?? '';

    $conn->beginTransaction();

    switch ($notificationType) {
        case 'meeting':
        case 'general_meeting':
        case 'progress_card':
        case 'overall_progress':
            // For Admin and Super Admin, mark notification as seen using admin views table
            if ($userRole === 'SuperAdmin' || $userRole === 'Super Admin' || $userRole === 'Admin') {
                $stmt = $conn->prepare("
                    INSERT INTO tbl_notification_admin_views (user_id, notification_id)
                    VALUES (?, ?)
                    ON DUPLICATE KEY UPDATE viewed_at = NOW()
                ");
                $stmt->execute([$userId, $notificationId]);

                $conn->commit();
                echo json_encode([
                    'status' => 'success',
                    'message' => 'Notification marked as seen successfully'
                ]);
                exit;
            }
            break;

        default:
            http_response_code(400);
            echo json_encode(['status' => 'error', 'message' => 'Invalid notification type']);
            exit;
    }

    $conn->commit();
}
```




```
try {
    // Get all class levels
    $stmt = $conn->prepare('SELECT level_id, level_name FROM tbl_student_levels');
    $stmt->execute();
    $levels = $stmt->fetchAll(PDO::FETCH_ASSOC);

    $scheduleResult = [];
    foreach ($levels as $level) {
        $level_id = $level['level_id'];
        $level_name = $level['level_name'];
        // Get schedule for this level
        $stmtSched = $conn->prepare('
            SELECT s.schedule_id, s.day_of_week, s.start_minutes, s.end_minutes, s.minutes, s.schedule_item_id,
            si.item_type, si.subject_id, si.subject_id_2, si.routine_id, si.routine_id_2,
            sub.subject_name, sub2.subject_name AS subject_name_2,
            r.routine_name, r2.routine_name AS routine_name_2
            FROM tbl_schedule s
            JOIN tbl_schedule_items si ON s.schedule_item_id = si.schedule_item_id
            LEFT JOIN tbl_subjects sub ON si.subject_id = sub.subject_id
            LEFT JOIN tbl_subjects sub2 ON si.subject_id_2 = sub2.subject_id
            LEFT JOIN tbl_routines r ON si.routine_id = r.routine_id
            LEFT JOIN tbl_routines r2 ON si.routine_id_2 = r2.routine_id
            WHERE s.level_id = ?
            ORDER BY FIELD(s.day_of_week, "Monday", "Tuesday", "Wednesday", "Thursday", "Friday"), s.start_minutes
        ');
        $stmtSched->execute([$level_id]);
        $schedules = $stmtSched->fetchAll(PDO::FETCH_ASSOC);
    }
}
```

```
$senderId = isset($payload['sender_id']) ? (int)$payload['sender_id'] : 0;
$receiverId = isset($payload['receiver_id']) ? (int)$payload['receiver_id'] : 0;
$messageText = isset($payload['message_text']) ? trim($payload['message_text']) : '';

if ($senderId <= 0 || $receiverId <= 0 || $messageText === '') {
    throw new Exception('Missing required fields: sender_id, receiver_id, message_text');
}

$stmt = $conn->prepare("INSERT INTO tbl_communication (sender_id, receiver_id, message_text, is_read, sent_at, is_archived)
    VALUES (:sender_id, :receiver_id, :message_text, 0, CURRENT_TIMESTAMP(), 0)");
$stmt->bindValue(':sender_id', $senderId, PDO::PARAM_INT);
$stmt->bindValue(':receiver_id', $receiverId, PDO::PARAM_INT);
$stmt->bindValue(':message_text', $messageText, PDO::PARAM_STR);
$stmt->execute();

$insertId = (int)$conn->lastInsertId();
// Retrieve the inserted row's timestamp
$stmt = $conn->prepare('SELECT sent_at FROM tbl_communication WHERE message_id = :id');
$stmt->bindValue(':id', $insertId, PDO::PARAM_INT);
$stmt->execute();
$row = $stmt->fetch(PDO::FETCH_ASSOC);

$response['success'] = true;
$response['data'] = [
    'message_id' => $insertId,
    'sent_at' => $row ? $row['sent_at'] : null,
];
```



```
try {
    $stmt = $conn->prepare("INSERT INTO tbl_comm_group_message (group_id, sender_id, message_text) VALUES (?, ?, ?)");
    $stmt->execute([$groupId, $senderId, $text]);
    $id = (int)$conn->lastInsertId();

    $row = $conn->prepare("SELECT group_message_id, group_id, sender_id, message_text, sent_at, is_unsent
    FROM tbl_comm_group_message WHERE group_message_id = ?");
    $row->execute([$id]);
    $msg = $row->fetch(PDO::FETCH_ASSOC);

    echo json_encode(['success' => true, 'data' => $msg]);
} catch (PDOException $e) {
    http_response_code(500);
    echo json_encode(['success' => false, 'error' => 'Database error', 'details' => $e->getMessage()]);
}
?>
```

```
try {
    // Fetch all advisories with their details
    $stmt = $pdo->prepare("
    SELECT
        a.advisory_id,
        a.level_id,
        a.lead_teacher_id,
        a.assistant_teacher_id,
        a.total_male,
        a.total_female,
        a.total_students,
        sl.level_name
    FROM tbl_advisory a
    LEFT JOIN tbl_student_levels sl ON a.level_id = sl.level_id
    ORDER BY a.advisory_id
    ");
    $stmt->execute();
    $advisories = $stmt->fetchAll(PDO::FETCH_ASSOC);

    // For each advisory, fetch only active assigned students
    foreach ($advisories as &$advisory) {
        $studentStmt = $pdo->prepare("
        SELECT
            sa.student_id,
            s.stud_firstname,
            s.stud_middlename,
            s.stud_lastname,
            s.parent_id
        FROM tbl_student_assigned sa
        JOIN tbl_students s ON sa.student_id = s.student_id
        WHERE sa.advisory_id = ? AND s.stud_school_status = 'Active'
        ");
        $studentStmt->execute([$advisory['advisory_id']]);
        $advisory['students'] = $studentStmt->fetchAll(PDO::FETCH_ASSOC);
    }
}
```



```

$input = json_decode(file_get_contents('php://input'), true);
$student_id = isset($input['student_id']) ? intval($input['student_id']) : null;
$activity_id = isset($input['activity_id']) ? intval($input['activity_id']) : null;
$visual_feedback_id = isset($input['visual_feedback_id']) ? intval($input['visual_feedback_id']) : null;

if (!$student_id || !$activity_id || !$visual_feedback_id) {
    echo json_encode(['status' => 'error', 'message' => 'Missing required fields.']);
    exit();
}

try {
    // Check if tracking record exists for this student/activity
    if (trackingRecordExists($conn, $student_id, $activity_id)) {
        // Update existing record
        if (updateTrackingRecord($conn, $student_id, $activity_id, $visual_feedback_id)) {
            echo json_encode(['status' => 'success', 'message' => 'Rating updated successfully.']);
        } else {
            echo json_encode(['status' => 'error', 'message' => 'Failed to update rating.']);
        }
    } else {
        // Insert new record
        if (insertTrackingRecord($conn, $student_id, $activity_id, $visual_feedback_id)) {
            echo json_encode(['status' => 'success', 'message' => 'Rating saved successfully.']);
        } else {
            echo json_encode(['status' => 'error', 'message' => 'Failed to save rating.']);
        }
    }
} catch (PDOException $e) {
    http_response_code(500);
    echo json_encode(['status' => 'error', 'message' => 'Database error', 'error' => $e->getMessage()]);
}

```

```

try {
    // 1. Get all active students in this advisory and session
    $stmt = $conn->prepare('SELECT s.student_id, s.stud_firstname, s.stud_middlename, s.stud_lastname
    FROM tbl_student_assigned sa
    JOIN tbl_students s ON sa.student_id = s.student_id
    WHERE sa.advisory_id = ? AND s.stud_schedule_class = ? AND s.stud_school_status = "Active"
    ORDER BY s.student_id');
    $stmt->execute([$advisory_id, $session_class]);
    $students = $stmt->fetchAll(PDO::FETCH_ASSOC);

    // 2. Get all activities for this advisory/subject
    $stmt = $conn->prepare('SELECT activity_id, activity_num, activity_name, activity_date, subject_id, quarter_id
    FROM tbl_activities
    WHERE advisory_id = ? AND subject_id = ? AND activity_status = "Active"
    ORDER BY activity_num DESC');
    $stmt->execute([$advisory_id, $subject_id]);
    $activities = $stmt->fetchAll(PDO::FETCH_ASSOC);

    // 3. Get all tracking (ratings) for these students and activities from appropriate level tables
    $studentIds = array_column($students, 'student_id');
    $activityIds = array_column($activities, 'activity_id');
    $tracking = getTrackingData($conn, $studentIds, $activityIds);

    echo json_encode([
        'status' => 'success',
        'students' => $students,
        'activities' => $activities,
        'tracking' => $tracking
    ]);
} catch (PDOException $e) {
    http_response_code(500);
    echo json_encode(['status' => 'error', 'message' => 'Database error', 'error' => $e->getMessage()]);
}

```



```

$insertedNames = [];
foreach ($student_ids as $sid) {
    file_put_contents(__DIR__ . '/debug_progress_card.txt', "DEBUG: Entered foreach for student $sid\n", FILE_APPEND);
    // 1. Get all subjects for this advisory (by level, matching dashboard logic)
    $subjects = [];
    $stmt = $conn->prepare('
        SELECT DISTINCT s.subject_id, s.subject_name
        FROM tbl_schedule sch
        JOIN tbl_schedule_items si ON sch.schedule_item_id = si.schedule_item_id
        JOIN tbl_subjects s ON si.subject_id = s.subject_id
        WHERE sch.level_id = (SELECT level_id FROM tbl_advisory WHERE advisory_id = ?)
    ');
    $stmt->execute([$advisory_id]);
    $subjectRows = $stmt->fetchAll(PDO::FETCH_ASSOC);
    foreach ($subjectRows as $row) {
        $subjects[] = $row['subject_id'];
    }
    if (empty($subjects)) {
        file_put_contents(__DIR__ . '/debug_progress_card.txt', "SKIP: No subjects for advisory $advisory_id\n", FILE_APPEND);
        continue;
    }
    // 2. Get all quarter feedback for this student/quarter/subjects
    $in = implode(',', array_fill(0, count($subjects), '?'));
    $params = array_merge([$sid, $quarter_id], $subjects);
    $sql = "SELECT subject_id, visual_feedback_id FROM tbl_quarter_feedback WHERE student_id = ? AND quarter_id = ? AND $in";
    $stmt2 = $conn->prepare($sql);
    $stmt2->execute($params);
    $feedbacks = $stmt2->fetchAll(PDO::FETCH_ASSOC);
    file_put_contents(__DIR__ . '/debug_progress_card.txt', "DEBUG: Feedbacks found: " . count($feedbacks) . ", Subject");
    if (count($feedbacks) < count($subjects)) {
        $feedbackSubjectIds = array_column($feedbacks, 'subject_id');
        $missingSubjects = array_diff($subjects, $feedbackSubjectIds);
        // Get missing subject names
        $missingNames = [];
        foreach ($subjectRows as $row) {
            if (in_array($row['subject_id'], $missingSubjects)) {

```



```
foreach ($student_ids as $sid) {
    if (count($feedbacks) < count($subjects)) {
        // Get missing subject names
        $missingNames = [];
        foreach ($subjectRows as $row) {
            if (in_array($row['subject_id'], $missingSubjects)) {
                $missingNames[] = $row['subject_name'];
            }
        }
        file_put_contents(__DIR__ . '/debug_progress_card.txt', "SKIP: Not enough feedbacks for student $sid, quarter $quarter");
        echo json_encode([
            "status" => "error",
            "message" => "Not all subjects have feedback. Missing feedback for: " . implode(', ', $missingNames),
            "missing_subjects" => $missingNames
        ]);
        exit;
    }
}

// 3. Get dynamic shape-to-score mapping from database
$shapeAvg = getDynamicShapeMapping($conn);
logShapeMapping("insert_progress_card", $shapeAvg, $conn);
// 4. Get shape for each feedback
$shapeScores = [];
foreach ($feedbacks as $fb) {
    $vf_id = $fb['visual_feedback_id'];
    $shapeRow = $conn->query("SELECT visual_feedback_shape FROM tbl_visual_feedback WHERE visual_feedback_id = " . intval($vf_id));
    if ($shapeRow && isset($shapeAvg[$shapeRow['visual_feedback_shape']])) {
        $shapeScores[] = $shapeAvg[$shapeRow['visual_feedback_shape']];
    } else {
        file_put_contents(__DIR__ . '/debug_progress_card.txt', "SKIP: No shape row for feedback visual_feedback_id $vf_id");
        continue;
    }
}
}
```

```
foreach ($student_ids as $sid) {
    $quarter_avg_score = round(($averageScore / 4.600) * 100, 3); // 4.600 is Heart (max)
    // 6. Determine quarter_visual_feedback_id (use averageScore, not percentage)
    $sql3 = "SELECT visual_feedback_id, visual_feedback_shape FROM tbl_visual_feedback WHERE ? >= min_score AND ? <= max_score";
    $stmt3 = $conn->prepare($sql3);
    $stmt3->execute([$averageScore, $averageScore]);
    $row = $stmt3->fetch(PDO::FETCH_ASSOC);
    if (!$row) {
        file_put_contents(__DIR__ . '/debug_progress_card.txt', "ERROR: No visual_feedback_id found for averageScore $averageScore");
        continue;
    }
    $quarter_visual_feedback_id = $row['visual_feedback_id'];
    $shape = $row['visual_feedback_shape'];
    // 7. Determine risk_id dynamically based on shape
    $risk_id = getRiskLevelForShape($shape, $conn);
    // Use user_id from POST data for finalized_by
    $finalized_by = $data['user_id'] ?? null;
    if (!$finalized_by) {
        file_put_contents(__DIR__ . '/debug_progress_card.txt', "ERROR: finalized_by (user_id) is null\n", FILE_APPEND);
        echo json_encode([
            "status" => "error",
            "message" => "User not logged in or user_id missing. Cannot finalize progress card.",
            "students" => []
        ]);
        exit;
    }
}
```



```
// Insert into tbl_users
$stmt = $conn->prepare("INSERT INTO tbl_users (
    user_role, user_pass, user_firstname, user_middlename, user_lastname, user_birthdate,
    user_email, user_contact_no, user_photo, user_status, is_new
) VALUES (
    ?, ?, ?, ?, ?, ?, ?, ?, ?, 'Active', 'Yes'
)");

$stmt->execute([
    $userRole,
    $hashedPassword,
    $firstname,
    $middlename,
    $lastname,
    $birthdate,
    $email,
    $contact,
    $photo
]);

$user_id = $conn->lastInsertId();
```



```
// 1. Get all students assigned to this advisory and session
$stmt = $conn->prepare('SELECT s.student_id FROM tbl_student_assigned sa JOIN tbl_students s ON
sa.student_id = s.student_id WHERE sa.advisory_id = ? AND s.stud_schedule_class = ?');
$stmt->execute([$advisory_id, $session_class]);
$students = $stmt->fetchAll(PDO::FETCH_COLUMN);
if (empty($students)) {
    echo json_encode(['status' => 'error', 'message' => 'No students found for this advisory and session.']);
    exit();
}

// 2. Get next activity_num for this advisory/subject/quarter
$stmt = $conn->prepare('SELECT COALESCE(MAX(activity_num), 0) + 1 AS next_activity FROM tbl_activities
WHERE advisory_id = ? AND subject_id = ? AND quarter_id = ? AND activity_status = "Active"');
$stmt->execute([$advisory_id, $subject_id, $quarter_id]);
$next_activity = $stmt->fetchColumn();
if (!$next_activity) $next_activity = 1;

// 3. Insert new activity into tbl_activities (now with quarter_id)
$insertActivity = $conn->prepare('INSERT INTO tbl_activities (advisory_id, subject_id, quarter_id,
activity_name, activity_date, activity_num, activity_status) VALUES (?, ?, ?, ?, ?, ?, "Active")');
$insertActivity->execute([$advisory_id, $subject_id, $quarter_id, $activity_name, $activity_date, $next_activity]);
$activity_id = $conn->lastInsertId();

// 4. Insert into appropriate tracking table for each student based on their level
foreach ($students as $student_id) {
    insertTrackingRecord($conn, $student_id, $activity_id);
}

// Resequence activity_num for all activities in this advisory, subject, and quarter
$stmt = $conn->prepare('SELECT activity_id FROM tbl_activities WHERE advisory_id = ? AND subject_id = ?
AND quarter_id = ? AND activity_status = "Active" ORDER BY activity_date ASC, activity_id ASC');
$stmt->execute([$advisory_id, $subject_id, $quarter_id]);
$ids = $stmt->fetchAll(PDO::FETCH_COLUMN);
$num = 1;
foreach ($ids as $id) {
    $update = $conn->prepare('UPDATE tbl_activities SET activity_num = ? WHERE activity_id = ?');
    $update->execute([$num++, $id]);
}
}
```

```
$sql4 = "INSERT INTO tbl_progress_cards (student_id, advisory_id, quarter_id, quarter_visual_feedback_id,
risk_id, quarter_avg_score, is_finalized, finalized_by, report_date) VALUES (?, ?, ?, ?, ?, ?, 1, ?,
NOW()) ON DUPLICATE KEY UPDATE quarter_visual_feedback_id=VALUES(quarter_visual_feedback_id),
risk_id=VALUES(risk_id), quarter_avg_score=VALUES(quarter_avg_score), is_finalized=1,
finalized_by=VALUES(finalized_by), report_date=VALUES(report_date)";
$stmt4 = $conn->prepare($sql4);
```



```
$stmt = $conn->prepare('
    SELECT
        pc.quarter_id,
        pc.quarter_visual_feedback_id,
        COUNT(*) as student_count
    FROM tbl_progress_cards pc
    JOIN tbl_students s ON pc.student_id = s.student_id
    JOIN tbl_student_assigned sa ON s.student_id = sa.student_id
    WHERE pc.advisory_id = ?
        AND s.parent_id IS NOT NULL
        AND s.stud_school_status = "Active"
        AND sa.advisory_id = ?
    GROUP BY pc.quarter_id, pc.quarter_visual_feedback_id
    ORDER BY pc.quarter_id, pc.quarter_visual_feedback_id
');
$stmt->execute([$advisory_id, $advisory_id]);
$results = $stmt->fetchAll(PDO::FETCH_ASSOC);
```




Appendix P

Endorsement letter

TO WHOM IT MAY CONCERN:

This is to formally endorse the capstone project entitled “Web-Based Students Childhood Monitoring System with Analytics for Learners Ville CDO” submitted by the following Bachelor of Science in Information Technology students:

KRISTOPHER V. DICHOS
PAUL ANDREW L. NERONA
CHRISTIAN EJAY H. CLARIÑO
JESSA H. DECENA
QUENNIELYN M. GALARPE
MARGARETH P. MANONGDO

The project focuses on developing an integrated, web-based system dedicated to the management and analysis of student data for Learners Ville CDO, with features for tracking attendance, developmental milestones, behavioral assessments, and facilitating secure, real-time parent-teacher communication. The system is designed specifically for early childhood education (ages 2-4) and leverages analytics for proactive identification of learning gaps, enabling timely intervention and support.

After careful evaluation, the project proposal has met the initial requirements for endorsement and is hereby recommended for implementation and continuation under the supervision of the assigned adviser.

We believe that this project will make a significant contribution to improving record management and information processing at Learners Ville CDO by digitizing manual processes, enabling real-time communication between teachers and parents, and providing predictive analytics to identify and address learning gaps early.

Endorsed by:

ALEXIS V. LAROSA. PhD, MMME, LpT
Capstone Project Adviser
College of Information Technology Education



Appendix Q

Functionality Test Results

Functionality Test Result of Super Admin

Item	Functionality test	Answer count		Percentage	
		Yes	No	Yes	No
	WEB-BASED STUDENT'S CHILDHOOD MONITORING SYSTEM WITH ANALYTICS FOR LEARNER'S VILLE CDO				
1	Can you access the Dashboard?	1	0	100%	0%
2	Can you see the header 'Super Admin Dashboard'?	1	0	100%	0%
3	Can you view active user statistics for Admins, Teachers, Parents, and Students?	1	0	100%	0%
4	Can you view class performance charts on the Dashboard?	1	0	100%	0%
5	Can you view upcoming meetings/events on the Dashboard?	1	0	100%	0%
6	Can you access quick actions (User Management, Schedule, Reports)?	1	0	100%	0%
7	Can you manage Users?	1	0	100%	0%
8	Can you add new Admin, Teacher, Parent, and Student accounts?	1	0	100%	0%
9	Can you update existing accounts?	1	0	100%	0%
10	Can you deactivate/reactivate accounts?	1	0	100%	0%
11	Can you manage Schedules?	1	0	100%	0%
12	Can you create and edit schedule items, such as subjects or routines?	1	0	100%	0%
13	Can you oversee all class schedules?	1	0	100%	0%
14	Can you generate institutional reports (performance,	1	0	100%	0%



	attendance)?				
15	Can you generate overall class progress analytics?	1	0	100%	0%
16	Can you download reports (PDF)?	1	0	100%	0%
17	Can you monitor Logs?	1	0	100%	0%
18	Can you access activity logs?	1	0	100%	0%
19	Can you audit user activity?	1	0	100%	0%
20	Can you view login history and actions performed?	1	0	100%	0%
21	Can you use Messaging?	1	0	100%	0%
22	Can you send announcements to Admins, Teachers, and Parents?	1	0	100%	0%

Functionality Test Result of Admin

Item	Functionality test	Answer count		Percentage	
		Yes	No	Yes	No
	WEB-BASED STUDENT'S CHILDHOOD MONITORING SYSTEM WITH ANALYTICS FOR LEARNER'S VILLE CDO				
1	Can you access the Dashboard?	1	0	100%	0%
2	Can you see the header 'Admin Dashboard'?	1	0	100%	0%
3	Can you view active user statistics for Admins, Teachers, Parents, and Students?	1	0	100%	0%
4	Can you view class performance progress charts?	1	0	100%	0%
5	Can you view upcoming meetings/events?	1	0	100%	0%
6	Can you access quick actions (User Management, Reports)?	1	0	100%	0%
7	Can you manage Users?	1	0	100%	0%
8	Can you add Teacher, Parent, and Student accounts?	1	0	100%	0%



9	Can you update user accounts?	1	0	100%	0%
10	Can you deactivate/reactivate accounts?	1	0	100%	0%
11	Can you manage Schedules?	1	0	100%	0%
12	Can you create and update schedules for classes/events?	1	0	100%	0%
13	Can you view all institutional schedules?	1	0	100%	0%
14	Can you generate student quarterly performance, per subject and attendance reports?	1	0	100%	0%
15	Can you download reports (PDF)?	1	0	100%	0%
16	Can you monitor Logs?	1	0	100%	0%
17	Can you access activity logs?	1	0	100%	0%
18	Can you review student and teacher activity history?	1	0	100%	0%
19	Can you use Messaging?	1	0	100%	0%
20	Can you send messages to Teachers and Parents?	1	0	100%	0%
21	Can you send announcements?	1	0	100%	0%

Functionality Test Result of Teacher

Item	Functionality test	Answer count		Percentage	
		Yes	No	Yes	No
	WEB-BASED STUDENT'S CHILDHOOD MONITORING SYSTEM WITH ANALYTICS FOR LEARNER'S VILLE CDO				
1	Can you access the Dashboard?	1	0	100%	0%
2	Can you see the header 'Teacher Dashboard'?	1	0	100%	0%
3	Can you view statistics for assigned students?	1	0	100%	0%
4	Can you view upcoming meetings/events?	1	0	100%	0%



5	Can you access quick links (Student Records, Reports)?	1	0	100%	0%
6	Can you manage Students?	1	0	100%	0%
7	Can you view student profiles and records?	1	0	100%	0%
8	Can you update academic and developmental progress?	1	0	100%	0%
9	Can you record Attendance?	1	0	100%	0%
10	Can you mark students Present/Absent?	1	0	100%	0%
11	Can you update attendance records in the class schedule?	1	0	100%	0%
12	Can you conduct Assessments?	1	0	100%	0%
13	Can you record academic assessments?	1	0	100%	0%
14	Can you finalize student performance per quarter?	1	0	100%	0%
15	Can you add comments for each student per quarter?	1	0	100%	0%
16	Can you download student progress (PDF)?	1	0	100%	0%
17	Can you use Messaging?	1	0	100%	0%
18	Can you send/receive messages to Parents or other accounts?	1	0	100%	0%
19	Can you send/receive messages to other accounts?	1	0	100%	0%

Functionality Test Result of Parent

Item	Functionality test	Answer count		Percentage	
	WEB-BASED STUDENT'S CHILDHOOD MONITORING SYSTEM WITH ANALYTICS FOR LEARNER'S VILLE CDO	Yes	No	Yes	No



1	Can you access the Dashboard?	1	0	100%	0%
2	Can you see the header 'Parent Dashboard'?	1	0	100%	0%
3	Can you view statistics for their child/children?	1	0	100%	0%
4	Can you view upcoming meetings/events?	1	0	100%	0%
5	Can you manage Profile?	1	0	100%	0%
6	Can you update your profile details (contact, email, etc.)?	1	0	100%	0%
7	Can you view Student Report Card?	1	0	100%	0%
8	Can you view your own child's academic performance?	1	0	100%	0%
9	Can you view child's milestone interpretation?	1	0	100%	0%
10	Can you view Student Details?	1	0	100%	0%
11	Can you view and edit sociodemographic information?	1	0	100%	0%
12	Can you access Calendar?	1	0	100%	0%
13	Can you view institutional calendar events?	1	0	100%	0%
14	Can you view scheduled meetings?	1	0	100%	0%
15	Can you use Messaging?	1	0	100%	0%
16	Can you send/receive messages to Teachers or other accounts?	1	0	100%	0%

**Total Summary**

User Role	Status	Summary
Super Admin	Passed	All functionalities work as expected.
Admin	Passed	All functionalities verified successfully.
Teacher	Passed	All functionalities working properly.
Parent	Passed	All functionalities verified successfully.



Appendix R

SUS Results and Computed Score

Web-based Students Childhood Monitoring System with Analytics for Learners Ville CDO	P1	P2	P3	P4	P5	P6	P7	P8	AVE
I think I would like to use this system frequently.	5	4	5	4	5	4	3	4	4.3
I found the system unnecessarily complex.	2	2	1	3	2	2	3	2	2.1
I thought the system was easy to use.	4	5	5	4	5	4	3	4	4.2
I think I would need technical support to use this system.	2	3	2	3	2	3	3	2	2.6
I found the various functions in this system were well integrated.	5	4	5	4	5	4	4	4	4.4
I thought there was too much inconsistency in this system.	2	2	1	2	2	2	3	2	2.0
I would imagine that most people would learn to use this system very quickly.	5	5	5	4	5	4	4	4	4.4
I found the system very cumbersome to use.	2	2	2	3	2	2	3	2	2.3
I felt confident using the system.	5	5	5	4	5	4	4	4	4.5
I needed to learn many things before using the system.	3	2	2	3	2	3	3	2	2.6
Raw X	19	18	20	15	20	15	17	15	17.4
Raw Y	14	14	16	12	13	14	11	14	13.5
AVERAGE	30.9								
SUS Score	77.25								
INTERPRETATION	GOOD								



Appendix S

ISO 25010 Evaluation Form

Title: Web-Based Student's Childhood Monitoring System with Analytics for Learner's Ville CDO

Purpose: This questionnaire aims to evaluate the system in terms of software quality based on the ISO 25010 model. The results will measure system performance and user satisfaction in the school management context.

Respondent Information

- **Name (Optional):** _____
- **Designation / Role:** ☐ Super Admin ☐ Admin ☐ Teacher ☐ Parent
- **Date:** _____

Instructions

Please rate each statement according to your level of agreement using the scale below:

Scale

Score	Verbal Interpretation
5	Strongly Agree – The feature is always present and satisfactory.
4	Agree – The feature is usually present and satisfactory.
3	Neutral – The feature is sometimes present but needs improvement.
2	Disagree – The feature is seldom present and unsatisfactory.
1	Strongly Disagree – The feature is not present at all.

**Part – Functionality**

Indicators	5	4	3	2	1
The system performs all required features as intended.	☐	☐	☐	☐	☐
The system produces accurate and reliable outputs (e.g., reports, records, analytics).	☐	☐	☐	☐	☐
The system correctly processes user inputs without errors.	☐	☐	☐	☐	☐
All functions work as described in the user interface.	☐	☐	☐	☐	☐
The system meets the operational needs of its intended users.	☐	☐	☐	☐	☐

Part – Usability

Indicators	5	4	3	2	1
The system interface is user-friendly and intuitive.	☐	☐	☐	☐	☐
Icons, labels, and menus are clear and easy to understand.	☐	☐	☐	☐	☐
It is easy to navigate between different modules or pages.	☐	☐	☐	☐	☐
The system's design and layout are visually appealing.	☐	☐	☐	☐	☐
The overall system is easy to learn and use.	☐	☐	☐	☐	☐

Part – Reliability

Indicators	5	4	3	2	1
The system operates smoothly without crashing or freezing.	☐	☐	☐	☐	☐
The system consistently performs as expected during use.	☐	☐	☐	☐	☐
The system properly saves data even after interruptions or errors.	☐	☐	☐	☐	☐



Error messages are clear and help users recover quickly.	☐	☐	☐	☐	☐
The system maintains data integrity and avoids duplication.	☐	☐	☐	☐	☐

Part – Efficiency

Indicators	5	4	3	2	1
The system loads quickly and responds to user commands without delay.	☐	☐	☐	☐	☐
The system efficiently processes and retrieves data.	☐	☐	☐	☐	☐
The system performs tasks with minimal lag time.	☐	☐	☐	☐	☐
The response time is acceptable even when multiple users are active.	☐	☐	☐	☐	☐
The system is optimized for performance across devices (desktop, tablet, mobile).	☐	☐	☐	☐	☐

Part – Satisfaction

Indicators	5	4	3	2	1
I am satisfied with the overall performance of the system.	☐	☐	☐	☐	☐
The system helps improve efficiency in completing my tasks.	☐	☐	☐	☐	☐
The system meets my expectations for its intended purpose.	☐	☐	☐	☐	☐
I would recommend the Learner's Ville CDO system to others.	☐	☐	☐	☐	☐
I am confident in using the system regularly for educational purposes.	☐	☐	☐	☐	☐



Appendix T

ISO 25010 Evaluation Results

Functionality

Indicators	R1	R2	R3	R4	R5	R6	R7	R8	Item Average
The system performs all required features as intended.	4	5	4	5	4	4	5	4	4.4
The system produces accurate and reliable outputs (e.g., reports, records, analytics).	4	5	5	4	4	4	5	4	4.4
The system correctly processes user inputs without errors.	5	4	4	5	4	4	5	4	4.4
All functions work as described in the user interface.	4	4	5	4	5	4	4	4	4.3
The system meets the operational needs of its intended users.	4	4	4	5	4	5	4	4	4.3
Average									4.36

Usability

Indicators	R1	R2	R3	R4	R5	R6	R7	R8	Item Average
The system interface is user-friendly and intuitive.	4	4	5	4	5	4	4	4	4.3
Icons, labels, and menus are clear and easy to understand.	4	4	4	5	4	4	5	4	4.3
It is easy to navigate between different modules or pages.	5	4	4	4	4	4	5	4	4.3



The system's design and layout are visually appealing.	4	5	4	4	4	5	4	4	4.3
The overall system is easy to learn and use.	4	4	4	4	5	4	5	4	4.3
Average									4.30

Reliability

Indicators	R1	R2	R3	R4	R5	R6	R7	R8	Item Average
The system operates smoothly without crashing or freezing.	4	5	4	4	4	4	5	4	4.3
The system consistently performs as expected during use.	4	4	4	5	4	4	4	5	4.3
The system properly saves data even after interruptions or errors.	4	4	4	4	4	5	4	4	4.1
Error messages are clear and help users recover quickly.	4	4	5	4	4	4	4	5	4.3
The system maintains data integrity and avoids duplication.	4	5	4	4	4	4	4	4	4.1
Average									4.22

Efficiency

Indicators	R1	R2	R3	R4	R5	R6	R7	R8	Item Average
The system loads quickly and responds to user commands without delay.	4	4	4	5	4	5	4	4	4.3
The system efficiently processes and retrieves data.	5	4	4	4	5	4	4	4	4.3



The system performs tasks with minimal lag time.	4	4	5	4	4	4	5	4	4.3
The response time is acceptable even when multiple users are active.	4	4	4	5	4	5	4	4	4.3
The system is optimized for performance across devices (desktop, tablet, mobile).	4	5	4	4	5	4	4	4	4.3
Average									4.32

Satisfaction

Indicators	R1	R2	R3	R4	R5	R6	R7	R8	Item Average
I am satisfied with the overall performance of the system.	4	5	4	4	5	4	4	5	4.4
The system helps improve efficiency in completing my tasks.	4	4	5	4	4	5	4	4	4.3
The system meets my expectations for its intended purpose.	4	4	4	5	4	4	5	4	4.3
I would recommend the Learner's Ville CDO system to others.	4	5	4	4	5	4	4	4	4.3
I am confident in using the system regularly for educational purposes.	5	4	4	4	4	5	4	4	4.3
Average									4.32



Appendix U

Visual Feedback Formula Reference

Shape Value Mapping Table

Shape	Min Score	Max Score	Range
 Heart	4.200	5.000	$4.200 \leq x \leq 5.000$
 Star	3.400	4.199	$3.400 \leq x \leq 4.199$
 Diamond	2.600	3.399	$2.600 \leq x \leq 3.399$
 Triangle	1.800	2.599	$1.800 \leq x \leq 2.599$
 Circle	1.000	1.799	$1.000 \leq x \leq 1.799$

Visual Feedback Scale Reference Table

Description	Shape	Scale Range	Remarks
Excellent	 Heart	4.200 – 5.000	Performance exceeds expectations
Very Good	 Star	3.400 – 4.199	Meets expectations, with minor improvements needed
Good	 Diamond	2.600 – 3.399	Acceptable, but improvement required
Need Help	 Triangle	1.800 – 2.599	Below expectations, needs guidance
Not Met	 Circle	1.000 – 1.799	Performance significantly



			below standard
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Quarterly Progress Example

Subject	Shape	Value	Calculation
Subject 1	♥	4.600	
Subject 2	★	3.7995	
Subject 3	★	3.7995	
Subject 4	◆	2.9995	
Subject 5	▲	2.1995	
Subject 6	●	1.3995	
Total		18.7975	Sum = 18.7975
Average		3.133	$18.7975 \div 6 = 3.133$
Percentage		68.109%	$(3.133 \div 4.6) \times 100 = 68.109\%$
Result Shape	◆	Diamond	3.133 falls in range 2.600–3.399

Overall Progress Example

Quarter	Percentage	Calculation
Q1	68.109%	From above example
Q2	70.000%	
Q3	65.000%	
Q4	72.000%	
Total	275.109%	Sum = 275.109%
Average	68.777%	$275.109 \div 4 = 68.777\%$



Scale Score	3.164	$(68.777 \div 100) \times 4.6 = 3.164$
Result Shape		Diamond

Subject Overall Progress Example

Math Subject for Student ID 6:

Quarter	Shape	Value	Calculation
Q1		3.7995	
Q2		3.7995	
Q3		2.9995	
Q4		3.7995	

Subject Overall Calculation:

Average: $(3.7995 + 3.7995 + 2.9995 + 3.7995) \div 4 = 3.600$

Percentage: $(3.600 \div 4.600) \times 100 = 78.261\%$

Final shape: 3.600 falls in range 3.400–4.199 →  Star

Student Quarter Feedback Pattern:

Subject	Q1	Q2	Q3	Q4
Basic Life Skills			-	-
Early Math			-	-
Literacy			-	-
Physical Activities			-	-
Science			-	-
Socio Emotional			-	-



Quarter 1 Calculation Example:

Values: $3.7995 + 2.1995 + 3.7995 + 3.7995 + 2.1995 + 4.600 = 20.398$

Average: $20.398 \div 6 = 3.400$

Percentage: $(3.400 \div 4.600) \times 100 = 73.913\%$

Shape: 3.400 falls exactly at Star boundary → ★ Star

Quarter 2 Calculation Example:

Values: $1.3995 + 1.3995 + 2.1995 + 1.3995 + 1.3995 + 1.3995 = 9.197$

Average: $9.197 \div 6 = 1.533$

Percentage: $(1.533 \div 4.600) \times 100 = 33.326\%$

Shape: 1.533 falls in range 1.000–1.799 → ● Circle